

CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION

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We study the minimum implementation cost of revising an installed policy while preserving operating performance after every state and date. A common randomized Markov continuation must serve all restarts. We construct a backward feasibility repair with an explicit cost modulus. A witness-driven form avoids an exact operating solve. It yields convergent global policy search for finite-state problems and an end-to-end approximation bound for continuous-state problems admitting certified total-variation aggregation. Exact rational spatial search and a verified McCormick comparator target the same constrained objective. A prospectively fixed eight-environment experiment reports all budget stops. On the original 42 continuous-state development configurations, whole-domain lotteries certify strict improvement over every deterministic policy in 27 cases. Nonlinear two-state lower bounds become positive, but longer-horizon gaps remain material. Independent checks distinguish arithmetic validity from approximation assumptions. The results identify when local operating guarantees can be converted into global revision certificates without changing the policy class.

KEYWORDS: Dynamic programming, Policy revision, Constrained Markov control, Global optimization, Verified computation.

1. INTRODUCTION

An organization may wish to improve an installed rule without replacing every decision it makes. Implementation cost then depends both on the decisions changed today and on the states visited under the revised rule tomorrow. A feasible revision must preserve the operating objective, whereas a cost-effective revision must economize on intervention along its own induced state distribution. These two requirements create a constrained dynamic problem. Pointwise selection of inexpensive changes need not solve it.

The relevant continuation must also be specified. Two histories that meet at the same date and state use the same continuation under a Markov rule. Allowing a different continuation for each initial condition or each remembered randomization can reduce costs, but changes the economic object. Conversely, requiring a deterministic rule can increase implementation cost even when randomization is admissible. We therefore take a common randomized Markov rule as the central choice variable. Performance is protected after every date and state; implementation cost is integrated under a stated initial distribution. A performance guarantee at every restart is not a claim of cost optimality at every restart.

The main contribution is a constructive connection between these requirements and a complete global search. A policy that violates every operating constraint by at most δ can be repaired backward without adding a persistent random seed or a history variable. The repair improves operating value at every state. Its additional implementation cost is bounded explicitly by a modulus that vanishes with δ . Strict discounting and a positive operating allowance provide the slack; no global duality assumption is used. This repair gives an error bound for rational policy lattices. For binary actions it also gives a stopping theorem for the interval Bellman branch-and-bound procedure used in the experiments. Unlike convergence to a necessary-action relaxation, the limiting object is the original common-continuation constrained optimum.

The witness-driven extension separates operating approximation error from policy-search error in one computable global bound. It supplies a positive-margin check and a constructive

policy without passing the exact optimum to the repair routine. A further perturbation result converts model error into operating and cost allowances; strict randomized separations also yield break-even administration charges. These statements extend the numerical method rather than changing its economic objective.

A second result transfers finite-state certificates to a continuous state space. Under a certified total-variation approximation of the transition kernel by a within-cell reset kernel, cell averaging maps an arbitrary measurable Markov policy to a finite-state policy. Lifting reverses this construction. Uniform primitive errors give an explicit operating-tolerance adjustment and a cost adjustment. The repair modulus controls the remaining difference between the relaxed and tightened finite-state optima. The theorem makes the approximation assumption operational and visible. It does not treat weak convergence of an atomic transition law as total-variation convergence, and it does not silently replace discrete shocks by continuous noise.

The numerical analysis separates three questions. First, does complete search return sound intervals for the common Markov problem, and how do those intervals compare with a direct global mathematical-programming formulation? Second, does randomization change the conclusion in the original continuous-state maintenance configurations? Third, can a nontrivial revision lower bound survive nonlinear, two-state dynamics and nonzero operating-witness error? Each experiment retains the same objective on the two sides of its interval. The eight new finite-state environments and both solver implementations were fixed before those environments were generated. The original 42 configurations are explicitly a development cohort, not a prospective test.

The resulting distinctions are economically consequential. In the development cohort, 27 whole-domain randomized deployments cost strictly less than a certified lower bound for every feasible deterministic policy under the same uniform initial law. This is a statement about the value of admissible randomization, not about a neural label attached to the installed rule. The nonlinear experiment changes the zero lower bound into a positive bound; its finest horizon-32 interval is $[8.8718, 11.8492]$. That interval is informative but not tight. In the prospective finite-state comparison, both methods face the same node allowance and wall-clock limit. The stronger linear relaxation often improves the lower endpoint, but longer horizons retain substantial residual intervals. The theory supplies completeness, not a claim that complete search is inexpensive.

1.1. *Relation to the literature*

Constrained Markov decision processes provide the natural starting point (Altman, 1999). Occupation measures are particularly effective when the constraints are expectations under a designated initial law. Our formulation additionally requires every restarted problem to meet its operating guarantee using one shared continuation. A separate occupation measure for each restart would need compatibility restrictions: at a shared date and state, its action probabilities cannot depend on which restart generated the occupancy. Dropping that restriction produces a different feasible class. Our finite formulation imposes it directly through a single policy variable for each date and state. This observation does not preclude occupation-measure methods; it specifies the coupling they must preserve in this problem.

Finite approximations to constrained control are not new. Saldi (2018) proves value convergence for discounted and average-cost constrained problems, and develops approximately optimal policies and rates under additional continuity conditions. Saldi, Linder, and Yüksel (2018) provides a systematic treatment of quantized stochastic control. Our transfer result concerns finite-horizon, all-restart constraints and combines the within-cell reset identity with an explicit Markov feasibility repair. It states the total-variation condition needed to average unrestricted measurable policies rather than assuming that a convenient policy sieve is automatically

dense in the constrained optimum. The theorem is not a replacement for those approximation theories; its additional object is the common-continuation restart constraint and its computable tolerance-to-cost modulus.

Primal–dual policy methods also solve constrained expected-return problems. Ding, Zhang, Basar, and Jovanovic (2020) establishes global convergence guarantees for natural policy-gradient methods in discounted constrained MDPs. Ni and Kamgarpour (2025) studies safe exploration under cumulative constraints. Those learning problems have information and feasibility requirements different from our known-model, all-restart certification problem. Safe baseline improvement (Laroche, Trichelair, and Tachet des Combes, 2019) and approximation-error analysis for fitted dynamic programming (Munos and Szepesvári, 2008, Scherrer, 2013) further distinguish model uncertainty, policy approximation, and optimization error. We claim no sample-efficiency comparison against methods given less information.

The global optimization ingredients are classical. Convex envelopes of factorable products originate in McCormick (1976); the numerical LP proposals here use a modern simplex implementation associated with Huangfu and Hall (2018). We do not attribute novelty to bilinear envelopes, best-bound search, or weak duality. The contribution is a common-policy economic formulation whose local approximation errors can be repaired into an admissible deployment, with an explicit global stopping bound. We use the direct McCormick formulation as a serious same-object comparator, not as a deliberately weak price portfolio. A rational residual correction turns proposed LP multipliers into lower bounds even when floating-point dual feasibility is imperfect. This separation of numerical proposal and proof follows the broader program of verified computation (Rump, 2010).

Semi-infinite optimization emphasizes the difference between checking finitely many constraints and controlling them on a region (Hettich and Kortanek, 1993). Our continuous-state certificates therefore include whole-cell inequalities and isolated switching points. Information-relaxation bounds supply another route to dynamic-programming lower and upper bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014); they are related to, but not identical with, the restart support functions used below. The support functions remain useful accelerators even though no finite price set is asserted to exhaust a global dual.

The economic motivation is adjustment rather than unconstrained operating optimization. Numerical economics has long emphasized model structure, approximation error, and state-space complexity (Judd, 1998, Brumm and Scheidegger, 2017). Empirical replacement models such as Rust (1987) discipline operating primitives with data. Our revision cost is instead the cost of departing from an installed decision rule, not a cost paid whenever the previous physical action changes. Adding physical switching costs would require including the previous action in the state. The present maintenance environments are designed, known-model economies. Their normalized operating and implementation units are not estimated monetary welfare, and their certificates are not estimates of empirical prevalence. The general convergence results, rather than a claimed calibration, carry the methodological argument.

2. THE ECONOMIC PROBLEM

Let X be a standard Borel space, $A = \{1, \dots, m\}$ a finite action set, and $t = 0, \dots, T$. Rewards $r_t(x, a)$, implementation costs $k_t(x, a)$, and terminal operating payoff $g(x)$ are bounded Borel functions. The transition kernels P_t^a are known Borel probability kernels. Write $|r_t| \leq R$, $|g| \leq G$, and $0 \leq k_t \leq K$. Unless otherwise indicated, $0 < \beta < 1$ and $\varepsilon > 0$. Define

$$T_t^a v(x) = r_t(x, a) + \beta P_t^a v(x), \quad V_T = g, \quad V_t = \max_{a \in A} T_t^a V_{t+1}. \quad (1)$$

Finiteness of A makes V_t Borel and permits a Borel maximizing selector, for example the lowest indexed maximizing action.

A randomized Markov policy is a sequence of Borel kernels $p_t(\cdot | x)$ on A . Its values satisfy

$$J_T^p = g, \quad J_t^p = \sum_a p_t(a | x) T_t^a J_{t+1}^p, \quad (2)$$

$$C_T^p = 0, \quad C_t^p = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}. \quad (3)$$

The same p_t is used by all histories reaching (t, x) . Randomization is newly drawn conditional on the current date and state; a persistent hidden coin indexing different future policies is not introduced.

For an initial probability law ν , the central object is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}_R(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}_R(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \forall t, x\}. \quad (4)$$

A deterministic policy restricts p_t to vertices of the simplex and defines K_D . An operating-optimal policy is feasible, so both classes are nonempty. On a finite state space, the product of policy simplexes is compact, J^p and C^p are continuous, and the feasible set is closed. The infimum in (4) is then attained. We use infimum, not an unproved function-space minimum, in the Borel-state formulation.

For an installed rule $a_t^0(x)$, the principal cost specification is

$$k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}, \quad w_t \geq 0. \quad (5)$$

The cost is charged for the action actually drawn. Its expectation under a lottery is not an indicator that the lottery's most probable action differs from the installed action. The unchanged installed policy has zero revision cost even when it is operating-infeasible. Consequently, dividing an optimality interval by its revision cost is undefined, not an informative relative-gap statistic. We report $(U - L)/U$ when $U > 0$, and report zero-cost exactness separately.

The uniform law in the development economy weights all normalized initial conditions equally. It is a numerical reference distribution, not an estimated stationary law. We also integrate exactly or outwardly under densities $2x$ and $2(1 - x)$. These calculations change the initial objective while leaving the all-state operating constraints and installed rule unchanged. A pointwise cost certificate would cover arbitrary initial laws, including atoms; an almost-everywhere cost enclosure need not.

3. FEASIBILITY REPAIR AND COMPLETE POLICY SEARCH

Set

$$D_C = K \sum_{s=0}^{T-1} (s+1) \beta^s, \quad (6)$$

$$D_J = \max_{1 \leq h \leq T} \left\{ 2R \sum_{s=0}^{h-1} (s+1) \beta^s + 2Gh\beta^h \right\}. \quad (7)$$

The maximum over remaining horizons in D_J matters: the terminal term $h\beta^h$ need not increase with h . Total variation between action distributions is one half of their ℓ_1 distance.

LEMMA 1—Policy perturbations: *If $\sup_{t,x} \text{TV}(p_t(\cdot | x), q_t(\cdot | x)) \leq \theta$, then*

$$\sup_{t,x} |J_t^p - J_t^q| \leq D_J \theta, \quad \sup_{t,x} |C_t^p - C_t^q| \leq D_C \theta.$$

PROOF: For costs, an action's remaining cost lies in $[0, K \sum_{s=0}^{h-1} \beta^s]$ at a state with h periods remaining. Changing its distribution changes the conditional expectation by at most the length of this interval times θ . Adding β times the next-date value difference and iterating gives $K \sum_{s=0}^{h-1} (s+1) \beta^s \theta$. For operating rewards use the interval length $2R$ for each stage and $2G$ for the terminal reward. A reward at distance s can be affected by $s+1$ action changes, and the terminal reward by h changes. Taking the maximum over h gives (7). The same conclusions follow by a finite-horizon coupling, but do not require one. *Q.E.D.*

THEOREM 1—Common-continuation repair: *Suppose $p \in \mathcal{F}_R(\varepsilon + \delta)$, $\delta \geq 0$. There is a Borel randomized Markov policy $\hat{p} \in \mathcal{F}_R(\varepsilon)$ with $J_t^{\hat{p}} \geq J_t^p$ for every (t, x) and*

$$\sup_{t,x} \text{TV}(\hat{p}_t, p_t) \leq \rho_\varepsilon(\delta) := \frac{\delta}{(1-\beta)\varepsilon + \delta}, \quad \int C_0^{\hat{p}} d\nu \leq \int C_0^p d\nu + D_C \rho_\varepsilon(\delta). \quad (8)$$

In particular,

$$0 \leq K_R(\nu, \varepsilon) - K_R(\nu, \varepsilon + \delta) \leq D_C \rho_\varepsilon(\delta). \quad (9)$$

PROOF: Proceed backward, with $J_T^{\hat{p}} = g$. Given a repaired continuation, put $q_a = T_t^a J_{t+1}^{\hat{p}}$, $q_p = \sum_a p_t(a | x) q_a$, $q_* = \max_a q_a$, and $b = V_t - \varepsilon$. By the induction hypothesis, $q_p \geq J_t^p \geq V_t - \varepsilon - \delta$. Also $J_{t+1}^{\hat{p}} \geq V_{t+1} - \varepsilon$ implies $q_* \geq V_t - \beta\varepsilon$. If $q_p \geq b$, retain p_t . Otherwise choose a measurable maximizing action a_* and set

$$\hat{p}_t = (1 - \alpha)p_t + \alpha e_{a_*}, \quad \alpha = \frac{b - q_p}{q_* - q_p}.$$

Writing $d = b - q_p \in (0, \delta]$, the denominator is at least $(1 - \beta)\varepsilon + d$. Thus $\alpha \leq \rho_\varepsilon(\delta)$ and $J_t^{\hat{p}} = b \geq q_p \geq J_t^p$. All operations are Borel; the division is used only where the denominator is strictly positive. The construction uses only t and x and the already fixed next-date continuation. Lemma 1 bounds cost. Apply the construction to arbitrarily near-minimizers at tolerance $\varepsilon + \delta$ and take their infimum to obtain (9). *Q.E.D.*

The repair is not a convex combination of whole policies. Such a combination, if selected once and remembered, would enlarge the information state. Here the mixture is selected anew at each state using one repaired continuation. The slack $(1 - \beta)\varepsilon$ explains why the theorem assumes strict discounting and a positive tolerance. No vanishing modulus at $\beta = 1$ or $\varepsilon = 0$ is asserted by this argument.

3.1. Certified search without an exact operating solve

The exact value in the preceding statements is not an oracle that the method must possess. A verified upper witness supplies the slack needed by the repair. This matters when operating approximation, rather than policy search alone, is expensive. Let $L_t \leq V_t \leq U_t$ be bounded

Borel witnesses with $U_T = g$. Define checked scalar bounds

$$\begin{aligned} e &\geq \max_{t < T} \|U_t - L_t\|_\infty, & \sigma &= (1 - \beta)\varepsilon - d, \\ d &\geq \max \left\{ 0, \sup_{t < T, x} \left[U_t(x) - \max_a T_t^a U_{t+1}(x) \right] \right\}. \end{aligned} \quad (10)$$

For the finite-state statement, let $m = |A|$, let $M \geq 1$ be an integer, and set $\theta_M = \min\{1, (m - 1)/M\}$. Lattice probabilities are multiples of $1/M$. The same primitive transition and reward operators occur in the witness check and the repair. Error in a different operator must first be included in its enclosure. A small residual is not, by itself, a certificate that $U \geq V$; the witness inequalities and terminal condition are also required.

THEOREM 2—Witness-driven repair and global error: *Suppose $\sigma > 0$ and $\delta \geq 0$. If a common randomized Markov policy satisfies $J_t^p \geq U_t - \varepsilon - \delta$ for all $t < T$, the backward construction with target $U_t - \varepsilon$ produces a common policy $\hat{p}$ such that*

$$J_t^{\hat{p}} \geq U_t - \varepsilon \geq V_t - \varepsilon, \quad J_t^{\hat{p}} \geq J_t^p, \quad \sup_{t, x} \text{TV}(\hat{p}_t, p_t) \leq \frac{\delta}{\sigma + \delta}. \quad (11)$$

Its additional initial implementation cost is at most $D_C \delta / (\sigma + \delta)$. On a finite state space, let $v_{M,L}$ be the smallest cost among rational-lattice policies satisfying $J_t^p \geq L_t - \varepsilon - D_J \theta_M$. Repair a minimizer using U , and let its cost be $U_{M,L,U}$. Then

$$\max\{0, v_{M,L} - D_C \theta_M\} \leq K_R(\nu, \varepsilon) \leq U_{M,L,U}, \quad (12)$$

$$U_{M,L,U} - \max\{0, v_{M,L} - D_C \theta_M\} \leq D_C \left\{ \theta_M + \frac{e + D_J \theta_M}{\sigma + e + D_J \theta_M} \right\}. \quad (13)$$

Thus $M \rightarrow \infty$, $e \rightarrow 0$, and $d \rightarrow 0$ give convergence to the actual constrained optimum using checked witnesses rather than an exact operating solve. All displayed errors and the stopping test are computable for rational represented finite-state primitives.

PROOF: Fix a repaired continuation and put $q_a = T_t^a J_{t+1}^{\hat{p}}$, $q_p = \sum_a p_t(a | x) q_a$, and $b = U_t - \varepsilon$. At the final decision date $J_T^{\hat{p}} = U_T$; at every earlier date the induction hypothesis gives $J_{t+1}^{\hat{p}} \geq U_{t+1} - \varepsilon$. Positivity and (10) imply

$$q_* := \max_a q_a \geq \max_a T_t^a U_{t+1} - \beta \varepsilon \geq U_t - d - \beta \varepsilon = b + \sigma.$$

The inductive improvement gives $q_p \geq J_t^p \geq b - \delta$. Where $q_p < b$, set $\alpha = (b - q_p)/(q_* - q_p)$ and mix p_t with the least-index maximizing action. For $z = b - q_p \in (0, \delta]$, one has $\alpha \leq z/(\sigma + z) \leq \delta/(\sigma + \delta)$. Otherwise leave p_t unchanged. This proves (11); the cost bound follows from Lemma 1. Finite action maximization and division on a strictly positive-denominator Borel set preserve measurability.

For the lattice statement, round any genuinely feasible policy as in Theorem 3. Its rounded value exceeds $V - \varepsilon - D_J \theta_M \geq L - \varepsilon - D_J \theta_M$, and its cost increases by at most $D_C \theta_M$. This proves the lower endpoint. A lattice minimizer is at worst $e + D_J \theta_M$ below the upper-witness target. Apply the first assertion with that value of δ to obtain the upper endpoint and (13). Nonnegative costs justify clipping the lower endpoint at zero. Q.E.D.

Failure of $\sigma > 0$ means that this particular witness pair has not certified the repair margin; it does not establish infeasibility of the original problem. Witness refinement and complete search have different roles in (13). Refining the witnesses alone does not close the remaining policy-search gap. Conversely, an increasingly fine policy lattice does not erase a fixed operating-witness error. The exact-value theorem is recovered by $L = U = V$ and $d = e = 0$.

3.2. A complete finite-state construction

Let X have n states. For an integer $M \geq 1$, let $\mathcal{P}_M$ be the policies whose action probabilities are multiples of $1/M$. Put $\theta_M = \min\{1, (m-1)/M\}$ and $\delta_M = D_J \theta_M$. Evaluate every candidate by the common Bellman equations and let

$$v_M = \min \left\{ \int C_0^p d\nu : p \in \mathcal{P}_M, J_t^p \geq V_t - \varepsilon - \delta_M \forall t \right\}.$$

For rational primitives these comparisons, values, and policy probabilities use rational arithmetic.

THEOREM 3—Convergence to the constrained optimum: *Repair a minimizer defining v_M and denote its feasible cost by U_M . Then*

$$L_M := \max\{0, v_M - D_C \theta_M\} \leq K_R(\nu, \varepsilon) \leq U_M, \quad U_M - L_M \leq D_C \{\theta_M + \rho_\varepsilon(\delta_M)\}. \quad (14)$$

Thus a prescribed positive absolute accuracy is reached at a finite M . At most $\binom{M+m-1}{m-1}^{nT}$ policies need be evaluated at that level. Running maxima of lower bounds and running minima of upper bounds give nested certificates without changing their validity.

PROOF: Round the first $m-1$ probabilities of any feasible policy down to multiples of $1/M$ and assign the remainder to the last action. The total-variation error is at most $(m-1)/M$, capped by one. Lemma 1 shows that the rounded policy belongs to the relaxed finite set and has cost at most the original cost plus $D_C \theta_M$. Hence $v_M - D_C \theta_M \leq K_R$. Theorem 1 applied to the minimizing grid policy gives $U_M \leq v_M + D_C \rho_\varepsilon(\delta_M)$. The weak-composition count gives the number of simplex points at each date and state. The displayed error tends to zero. *Q.E.D.*

The theorem controls distance to (4), not distance to a one-step relaxation. Its count is exponential in $nT(m-1)$; it is a completeness construction, not a polynomial-time claim. The deterministic finite-state problem can instead be solved by evaluating its m^{nT} vertices with the original tolerance. The randomized repair is neither required nor valid as a deterministic-preserving operation.

3.3. The executed spatial search

For binary actions, write p_{ti} for the probability of action 1. A policy box $Q = \prod_{ti} [\ell_{ti}, u_{ti}]$ retains the same coordinates across all restarts. Backward interval Bellman evaluation gives operating upper values $\bar{J}$ and unconstrained implementation lower values $\underline{C}$ for that box. Necessary operating tests can contract its coordinate intervals: if q_0^+, q_1^+ are next-date operating upper values, an admissible p must satisfy

$$(1-p)q_0^+ + pq_1^+ \geq V_t(i) - \varepsilon. \quad (15)$$

Intersect this half-interval with $[\ell_{ti}, u_{ti}]$. Empty intersections or inconsistent directed operating enclosures remove the box. Every removal is justified by a necessary inequality. Feasible policies are never removed because a sufficient candidate check fails.

The remaining cost lower bound is $\int \underline{C}_0 d\nu$. Repairing the midpoint of the contracted box produces a feasible upper policy. The algorithm selects a live box with the smallest lower bound and bisects a coordinate maximizing $\beta^t(u_{ti} - \ell_{ti})$. It preserves both children and every live leaf. A global lower endpoint is the minimum of all live or cost-pruned leaf lower bounds and the incumbent upper cost. It stops when their difference is no greater than the requested tolerance, or reports the remaining interval when the preassigned budget is exhausted.

PROPOSITION 1—Stopping bound for interval Bellman search: *Let $w = \max_{ti}(u_{ti} - \ell_{ti})$ for a surviving policy box. Its midpoint-repair cost minus its interval Bellman lower bound is at most*

$$H(w) = D_C\{w + \rho_\varepsilon(D_J w)\}. \quad (16)$$

Without a node or time cap, the best-bound bisection procedure reaches every prescribed positive gap η in finitely many steps. In particular, choose $w_ > 0$ with $H(w_*) \leq \eta$. A sufficient worst-case bisection depth is*

$$d \left\lceil \log_2 \frac{1}{\beta^{T-1} w_*} \right\rceil, \quad d = nT, \quad (17)$$

with the ceiling replaced by zero when its argument is at most one. The associated full binary-tree node bound is exponential in this depth. Adding valid McCormick lower bounds cannot invalidate the stopping guarantee.

PROOF: With only rectangular probability restrictions, ordinary Bellman maximization produces the maximum operating value simultaneously at every state. It is a common Markov policy, not a family of incompatible histories. A surviving contracted box has its operating upper value at least $V_t - \varepsilon$ at every date and state. Each such maximizing policy differs from the midpoint policy by total variation at most w . Lemma 1 therefore makes the midpoint $D_J w$ -approximately feasible. The same rectangular Bellman argument for minimum cost gives a lower bound no smaller than midpoint cost minus $D_C w$. Repair and (8) imply (16). A box satisfying $H(w) \leq \eta$ cannot need refinement while the global gap exceeds η : its repaired cost is already an incumbent candidate, so its lower bound is within η of the global incumbent or higher. On any still-refined path, weighted longest-side bisection makes every physical width at most w_* by the depth in (17). There are finitely many nodes up to that depth. Valid stronger lower bounds only remove or improve boxes. The supplement details the effect of the necessary contractions; they shrink the rectangles and preserve every feasible point. *Q.E.D.*

This is a convergence theorem for the implemented search rule, not an inference from favorable outcomes. It also explains why a 60-second run can end with a large gap: a finite worst-case bound can be prohibitively large. The numerical results report that distinction directly.

4. A SINGLE ERROR BUDGET FOR CONTINUOUS-STATE APPROXIMATION

A finite-state theorem alone does not justify replacing a continuous model by a grid. The following sufficient conditions permit such a transfer without restricting the original policy to be piecewise constant.

ASSUMPTION 1—Certified reset approximation: *For a finite Borel partition $(E_i)_{i=1}^{n_h}$, let μ_i be probability measures supported on E_i and write $\nu = \sum_i \nu_i \mu_i$. There are cellwise rewards $r_{h,t}$, costs $k_{h,t} \in [0, K]$, and terminal values g_h , and a kernel*

$$Q_{h,t}^a(x, dy) = \sum_j P_{h,t}^a(i, j) \mu_j(dy), \quad x \in E_i,$$

with certified uniform errors d_r, d_k, d_g and α_h satisfying

$$|r - r_h| \leq d_r, \quad |k - k_h| \leq d_k, \quad |g - g_h| \leq d_g, \quad |(P_t^a - Q_{h,t}^a)f(x)| \leq \alpha_h \|f\|_\infty$$

for every bounded Borel f . The approximation rewards obey the same bounds R, G as the original rewards, or R, G are enlarged to cover both models. All finite transition entries and error bounds are computable with verified directions.

The last inequality uses the operator convention for total variation; it is not the half- ℓ_1 convention for action probabilities above. Define $M_{J,t} = R \sum_{s=0}^{T-t-1} \beta^s + G \beta^{T-t}$ and $M_{C,t} = K \sum_{s=0}^{T-t-1} \beta^s$. Set

$$a_T = d_g, \quad a_t = d_r + \beta a_{t+1} + \beta \alpha_h M_{J,t+1}, \quad (18)$$

$$b_T = 0, \quad b_t = d_k + \beta b_{t+1} + \beta \alpha_h M_{C,t+1}, \quad (19)$$

$$a_h = \max_t a_t, \quad b_h = b_0.$$

Let $v_h(s)$ be the optimum of the finite common-Markov model at operating allowance s and initial masses (ν_i) .

THEOREM 4—End-to-end transfer: *Under Assumption 1, if $\varepsilon > 2a_h$, then*

$$v_h(\varepsilon + 2a_h) - b_h \leq K_R(\nu, \varepsilon) \leq v_h(\varepsilon - 2a_h) + b_h. \quad (20)$$

Suppose finite computation returns intervals $[L_+, U_+]$ and $[L_-, U_-]$ for these relaxed and tightened optima, of widths at most η_+ and η_- . A valid global interval is

$$[\max\{0, L_+ - b_h\}, U_- + b_h], \quad (21)$$

and its width is at most

$$\eta_+ + \eta_- + 2b_h + D_C \frac{4a_h}{(1 - \beta)(\varepsilon - 2a_h) + 4a_h}. \quad (22)$$

If all certified primitive errors tend to zero and both finite-solver widths tend to zero, this interval converges to the original measurable-policy optimum. Its measured width supplies a termination test for the actual economic objective.

PROOF: Fix any measurable Markov policy p , not necessarily cellwise constant. Backward subtraction of its original and reset-model Bellman equations gives (18)–(19). Bellman comparison gives the same operating error for optimal values. In the reset model, averaging the policy within E_i , $\bar{p}_t(a | i) = \int p_t(a | x) \mu_i(dx)$, averages its value exactly; after every transition the within-cell law is again μ_j , and stage rewards and costs are cellwise constant. This identity follows backward, including the common continuation. An original feasible policy has reset

operating value at least the reset optimum minus $\varepsilon + 2a_h$ pointwise, hence also after cell averaging; its finite-model cost differs from the original initial cost by at most b_h . Taking an infimum gives the lower inequality. Conversely, lift a finite policy feasible at $\varepsilon - 2a_h$ by using its action probabilities throughout each cell. The reset value is constant in each cell; the two operating errors make the lifted original policy feasible, and its cost differs by at most b_h . This proves the upper inequality. Finally apply Theorem 1 in the finite model with target allowance $\varepsilon - 2a_h$ and violation $4a_h$, then add the two finite-computation widths and the cost errors. *Q.E.D.*

4.1. Where the transfer theorem applies

On a compact domain, a transition density with computable uniform continuity in the source variable and an L_1 -approximable density in the destination variable can furnish the reset approximation. The cell reference measures must also represent the initial law as in Assumption 1. For example, with Lebesgue-uniform cell measures and a uniform initial law, a uniformly Lipschitz transition density gives a computable $\alpha_h = O(h)$ on a shape-regular mesh. Cellwise operating and cost approximations give $d_r, d_k, d_g = O(h)$ when their moduli are Lipschitz. Aligned discontinuities in an installed rule can be retained as exact partition boundaries. In this case (22) is $O(h + \eta_+ + \eta_-)$ at fixed positive ε and $\beta < 1$, with explicitly displayed constants. The number of cells still scales as $h^{-\dim X}$; the theorem does not remove the curse of dimensionality.

The primary affine-shock model and the nonlinear two-state model below have atomic transition laws. Smearing their successor atoms into cells generally leaves the total-variation error bounded away from zero. We therefore do *not* apply Theorem 4 to those models by asserting a small mesh. Their reported intervals are direct whole-domain bounds for the unchanged atomic models. They remain valid without a convergence claim for that particular spatial representation. This distinction prevents the approximation theorem from being used as an unsupported explanation for closing an atomic-model gap.

5. CERTIFICATES THAT ACCELERATE THE GLOBAL PROBLEM

The complete search and the transfer theorem provide the convergence argument. The inherited one-sided certificates are useful accelerators and diagnostic tools, but are not substituted for that argument.

5.1. Deterministic outer restrictions and measurability

Let $L_t \leq V_t \leq U_t$ be bounded Borel operating witnesses, including the terminal date. Every all-restart feasible deterministic policy must select an action in

$$\mathcal{N}_t^{L,U}(x) = \{a : L_t(x) - T_t^a U_{t+1}(x) \leq \varepsilon\}. \quad (23)$$

Indeed, its operating continuation cannot exceed $V_{t+1} \leq U_{t+1}$. The set is nonempty because an operating-optimal action satisfies the test. Its graph is Borel, since it is a finite union of Borel sublevel sets. Backward minimization of $k_t + \beta P_t^a B_{t+1}$ over this set yields a Borel lower function; the lowest-index minimizer is a Borel selector. Thus the general-state recursion does not rely on an implicit measurable-selection assertion.

This is a necessary-action outer class, not the original constrained problem. A selected continuation can accumulate losses that violate later operating requirements. Equality between its integrated cost lower bound and a separately feasible policy's cost certifies $K_D(\nu, \varepsilon)$. Equality of the full functions certifies more. Tightening L, U alone does not force equality; complete search is what controls the remaining structural gap.

5.2. Randomized support floors and a connected local program

For $\lambda \geq 0$, a verified supersolution W_t^λ of

$$W_T^\lambda \geq \lambda g, \quad W_t^\lambda \geq \max_a \{\lambda r_t - k_t + \beta P_t^a W_{t+1}^\lambda\}$$

implies $C_t^p \geq \lambda(V_t - \varepsilon) - W_t^\lambda$ for every feasible policy, including randomized Markov policies. Define a floor F_t by the maximum of zero and finitely many such terms, using L_t in place of V_t when necessary. Given a valid next-date lower function B_{t+1} , define

$$\mathcal{D}_t(x) = \left\{ p \in \Delta(A) : \sum_a p_a T_t^a U_{t+1}(x) \geq L_t(x) - \varepsilon \right\}.$$

$$B_t(x) = \max \left\{ F_t(x), \min_{p \in \mathcal{D}_t(x)} \sum_a p_a \{k_t(x, a) + \beta P_t^a B_{t+1}(x)\} \right\}, \quad B_T = 0. \quad (24)$$

The feasible action distribution of any admissible policy belongs to the local constraint, and its continuation cost dominates B_{t+1} . Induction proves that (24) is a global randomized cost lower bound. It does not prove that the local optimizer can be combined into a feasible policy or that finite support prices close a global duality gap.

For three actions and one local operating inequality, the local polytope has only pure feasible vertices and two-action boundary mixtures. Our new primary calculation evaluates these vertices with exact rational arithmetic on directed whole-cell bounds. Unlike the earlier diagnostic with a zero global floor, it uses the nonzero F_t at every date and is paired with a feasible policy for the same uniform-initial randomized objective. It improves that lower endpoint in 23 of the 42 configurations. The prior adaptive local-LP theorem and all of its experimental records remain in the retained supplement; their target is still the local relaxation, not (4).

5.3. Whole-domain randomized deployments

A simple backward construction supplies feasible upper policies without changing the atomic model. Given a feasible next-date value J_{t+1} , write $q_0 = T_t^{a_0} J_{t+1}$, $q_* = \max_a T_t^a J_{t+1}$, and $b = V_t - \varepsilon$. If $q_0 \geq b$, retain the installed action. Otherwise mix it with a lowest-index maximizing action with probability

$$p_t^{\text{change}}(x) = \frac{b(x) - q_0(x)}{q_*(x) - q_0(x)}. \quad (25)$$

The denominator is positive wherever this expression is used. The resulting operating value is $J_t = \max\{q_0, b\}$ and is feasible at every state. In the piecewise-affine model, J_t remains piecewise affine while the mixing probability can be a ratio of affine functions. Cost is therefore enclosed rather than falsely represented as an exact piecewise-affine function.

The verifier includes isolated switching points in every operating identity. Directed cost boxes integrate against the three absolutely continuous initial laws. All affine transition slopes are strictly positive, so a finite set of switching points and its finite-horizon preimages have zero probability under these laws. That fact justifies almost-everywhere cost enclosures; it does not justify ignoring knots in a restart operating constraint or using the same cost enclosure for a Dirac initial law.

5.4. A verified direct mathematical-programming comparator

The direct finite-state formulation minimizes $\sum_i \nu_i C_{0i}$ subject to (2)–(3), $J_{ti} \geq V_{ti} - \varepsilon$, and the common policy simplexes. The constraints are bilinear in policy probabilities and continuation values. A McCormick relaxation on each probability box provides a classical global lower program. The baseline uses the same incumbent repair, contraction rule, box selection, node allowance, initial law, and all-restart constraints as interval Bellman search. The additional work is the stronger LP bound and an LP-proposed feasible candidate after repair.

Floating-point LP status is not a certificate. For a minimization relaxation $c'z$ subject to $Az \leq b$ and $\ell \leq z \leq u$, every rational vector $\lambda \geq 0$ gives

$$\mathcal{L}(\lambda) = -\lambda'b + \sum_j \min\{(c + A'\lambda)_j \ell_j, (c + A'\lambda)_j u_j\} \leq \min c'z. \quad (26)$$

This follows by adding $\lambda'(Az - b) \leq 0$ and minimizing the resulting affine function over the box. Rounded nonnegative solver marginals are evaluated in (26) using exact rational arithmetic. A failed LP falls back to the interval Bellman lower bound; a floating infeasibility declaration never removes the policy box. A separate verifier rebuilds every relaxation and every search-tree partition from the primitive model.

6. NUMERICAL EVIDENCE

The experiments keep model class, information, initial distribution, and operating allowance explicit. The supplement reports every configuration; no failed or budget-stopped case is removed. The computation report records proof-object sizes, memory, wall time, the exact environment, and the boundary between independent arithmetic checks and shared-code validation.

6.1. Prospectively fixed common-Markov problems

Before generating eight new environments, we committed their seeds, joint parameter changes, two solver hashes, node cap 4,096, wall-clock cap 60 seconds per solver and case, and absolute stopping allowance 10^{-3} . The freeze is `commit 13fd3ee`; it follows development seed 390001 and precedes all eight evaluated seeds. Both solvers see the same known model. The two-state models represent low and high condition; four-state models use a pair of condition indicators with interaction terms in their transition probabilities. Horizons are 3, 4, 8, and 12; discount factors are 0.8 or 0.95; allowances are 0.01 or 0.05; operating prices, transition persistence, intervention effects, and initial masses vary jointly. The exact formulas and resulting rational matrices are in the supplement and proof objects.

Table I contains all eight pairs, including unfinished searches. A positive absolute-gap certificate is not necessarily a small relative-gap certificate when the optimum is small. We therefore report both. The linear relaxation often improves the lower endpoint substantially, but costs more per node. The four-state longer-horizon cases expose the combinatorial burden: the remaining interval is not hidden by a success count. The comparison does not establish that interval Bellman search dominates direct global programming. Instead it identifies the value and cost of the stronger relaxation under matched information and caps.

6.2. The unchanged continuous-state development cohort

The original maintenance model has $x \in [0, 1]$, three actions, two shocks with probabilities $3/5$ and $2/5$, $\beta = 19/20$, stage reward $x - c_a$ with $c = (0, 3/20, 9/20)$, terminal payoff $x/2$,

TABLE I
SAME-OBJECT COMPARISON ON THE PROSPECTIVELY FIXED ENVIRONMENTS

States	T	Method	Lower	Upper	Gap/upper (%)	Seconds
2	3	I	2.8414	2.8425	0.03	0.99
2	3	M	2.8414	2.8425	0.03	1.77
2	4	I	0.0000	0.0000	0.00	0.00
2	4	M	0.0000	0.0000	0.00	0.38
2	8	I	7.2534	9.0046	19.45	17.19
2	8	M	8.5290	9.0046	5.28	60.41
2	12	I	7.1419	7.5553	5.47	16.32
2	12	M	7.4446	7.5553	1.46	60.40
4	3	I	2.6157	2.6217	0.23	10.82
4	3	M	2.6206	2.6217	0.04	35.99
4	4	I	0.0233	0.0244	4.00	0.47
4	4	M	0.0235	0.0244	3.13	2.60
4	8	I	4.6631	7.8481	40.58	60.04
4	8	M	6.1990	7.8481	21.01	60.41
4	12	I	4.8936	6.6439	26.34	60.04
4	12	M	5.8606	6.6439	11.79	60.43

I: interval Bellman search. M: verified McCormick search. Bounds concern the identical common randomized Markov objective for each pair. Reported times include model construction and the operating solve; the search's 60-second cap is checked between iterations, not by interrupting a proof-producing node. A zero upper cost is an exact zero-cost case.

Full stop reasons, node counts, and memory appear in the supplement.

and revision weight $1 + x$. Its two affine successors for actions 0, 1, and 2 respectively are

$$(3x/4, 3x/4 + 1/20), \quad (17x/20 + 1/10, 17x/20 + 7/50), \quad (x/10 + 4/5, x/10 + 17/20).$$

The seven installed rules, horizons 4, 8, and 12, and allowances 0.01, 0.05 are unchanged. Neural and spline rules are fixed computational incumbents, not estimated economic policies. Exact operating values are inexpensive in this model and their computation is charged. We do not use this cohort to claim performance when operating approximation itself is difficult.

The inherited deterministic comparison has 12 zero-cost and 12 positive-cost integrated equalities. Ten of the positive-cost equalities occur at horizon 4, two at horizon 8, and none at horizon 12. The maximum deterministic relative gaps at each horizon are reported alongside absolute gaps in the supplement. In the horizon-8 neural31003 case at $\varepsilon = 0.05$, the two initial cost functions are identical pointwise, but the arrays differ at later restart dates. Thus initial equality does not imply equality at every restart. The new diagnostic records the full rational disagreement locus.

The lotteries in (25) certify strict improvement over *every* feasible deterministic policy in 27 of the 42 uniform-initial configurations (Table II). The horizon-12 defer case at allowance 0.01, for example, has a randomized interval $[15.3172, 15.4670]$ and a positive deterministic separation. This addresses the economically relevant policy-class difference directly, without substituting a fixed-restart history-conditioned frontier. The 12 zero-cost cases are already optimal by feasibility and nonnegativity and are not counted as nontrivial search successes. All 30 positive-cost randomized intervals retain a reported nonzero width; the narrowest and widest relative intervals are distinguishable in the full table.

Changing the initial density to $2x$ yields 28 strict deterministic separations; changing it to $2(1 - x)$ yields 27. These are sensitivity results on the same developed environments, not independent observations. The connected local program improves 23 uniform lower endpoints,

TABLE II
COMMON RANDOMIZED MARKOV CERTIFICATES ON THE DEVELOPMENT COHORT

T	Cases	Zero-cost exact	Strict gain over K_D	Improved lower	Maximum gap/upper (%)
4	14	4	10	8	14.61
8	14	4	8	7	32.29
12	14	4	9	8	34.23

Strict gain means that a verified randomized upper cost is below the deterministic lower bound for the same initial law and constraints. It does not mean that the randomized optimum is attained by that candidate. Improved lower compares the connected local-support bound with the prior randomized support lower bound. All 42 uniform-initial rows and both nonuniform sensitivity tables are retained.

TABLE III
SELECTED NONLINEAR COMMON-MARKOV INTERVALS

T	Mesh per axis	Lower	Upper	Gap/upper (%)	Independent check
8	64	0.9438	3.9348	76.01	full
16	128	4.1908	7.6644	45.32	not independent
32	64	6.5389	16.7884	61.05	full
32	128	8.1264	12.5226	35.11	full R40
32	256	8.8718	11.8492	25.13	not independent

Uniform initial law on the square and operating allowance $1/2$. Inherited full independent scalar checks cover $(T, N) = (8, 64)$ and $(32, 64)$. R40 additionally replays the complete $(32, 128)$ object, including the candidate upper cost and all-restart operating bound. The $(32, 256)$ result retains directed construction and structural checks, not the independent label of adjacent rows.

including the longer-horizon cases; it thus affects the primary global interval rather than a disconnected local diagnostic. It still does not establish global strong duality or close every interval.

6.3. Nonlinear two-state bounds

The retained nonlinear model has $(x, y) \in [0, 1]^2$, four actions, bilinear transitions, two shocks, reward $(x + y)/2 - c_a$, terminal payoff $(x + y)/4$, and revision weight $1 + (x + y)/2$. Its exact rational primitive specification is reproduced in the supplement. Whole-box operating witnesses and support supersolutions use outward arithmetic. The new lower construction uses 11 global support prices and five nonnegative local multipliers, including zero, at every date. It is paired with exactly the retained feasible upper policies. We run no new upper-portfolio competition and do not compare unequal proposal success fractions as evidence of optimality.

The old zero lower endpoint was not forced by the definition of revision cost: the operating constraints can force departures from the installed rule. Nonzero support supersolutions now detect that restriction. At $T = 32, N = 256$, the interval is $[8.8718, 11.8492]$, with relative width 25.13%. This is a genuine improvement over $[0, 11.8492]$, but it is not a near-exact optimum or a high-dimensional scalability result. Both endpoints and the all-restart operating inequality are independently rebuilt at $T = 32, N = 64$, and the R40 audit extends this to the complete $T = 32, N = 128$ object. Memory for the retained support arrays grows as $8N^2(T + 1)(11 + 4)$ bytes before temporary arrays and interpreter overhead; the machine record reports the actual payload and process high-water mark.

The allowance $1/2$ belongs to the inherited exploratory nonlinear design. It is not made comparable to 0.01 simply by normalizing state variables: the operating scale and remaining

horizon matter. We additionally compute lower-bound diagnostics at $\varepsilon = 0.05$ on four stated configurations. None is paired with a policy certified only at $1/2$. Without a newly feasible upper policy those tighter-tolerance rows remain lower-bound-only results. A failed sufficient box certificate is labeled uncertified, not infeasible. A box crossing an installed-rule threshold can charge the upper revision weight conservatively; it does not assert that every realized state in that box changes action.

7. VERIFICATION, ECONOMIC INTERPRETATION, AND LIMITS

A certificate has three distinct burdens. Its mathematical reduction must preserve the target policy class. Its finite arithmetic must check the asserted inequalities with the correct direction. Its publication must correspond to the checked objects. Hashes address the third burden, not the first two. Independent arithmetic addresses part of the second burden and can expose a model-contract error, but is not a proof that every economic assumption is appropriate.

The finite-state verifier uses SymPy rationals and imports neither search implementation nor the optimizer. It rebuilds the common Markov equations, primitive contracts, every policy-box split, every necessary contraction, the LP residual bounds, and the global leaf minimum. The continuous-state verifier independently checks the new lotteries, whole-cell costs, support supersolutions, and local primal vertices, using the prior independent primitive parser rather than the constructor's piecewise-affine operations. The nonlinear verifier uses symbolic rational corner geometry and arbitrary-precision scalar integer induction instead of the constructor's vectorized box operations. It checks two complete representative objects, not merely selected rows. The verification report specifies precisely which objects passed; it never substitutes cross-run equality or a mutation test for an inequality proof.

The synthesis preserves all prior scientific material. Deterministic bounds, history-conditioned frontiers, the two-period analytic randomization example, adaptive local-LP analysis, 66 economic sensitivities, failed runs, and the stopped-control derivations remain available in full. The supplement attaches the prior article and technical supplement verbatim, with a content map. The older 349-page historical annex remains unchanged as an archival object rather than a prerequisite for reading this article. The stopped consumption–portfolio problem is not relabeled solved: its inherited full-domain bound of approximately 7.1818 does not meet its 0.01 target. A killed Bellman comparison supports witness propagation but supplies neither missing action coverage nor a diffusion-discretization error bound (Kushner and Dupuis, 2001).

The economic conclusion established by the new primary calculation is precise. When randomized action assignment is admissible and costs attach to realized departures, common Markov lotteries can strictly reduce the minimum implementation cost relative to every deterministic feasible rule. This conclusion holds on the certified configurations and initial laws. It does not estimate how frequently organizations permit such lotteries, how costly they are to administer, or the monetary welfare effect of a real deployment. Fixed randomization overhead, parameter uncertainty, and empirical calibration would require additional primitives and certificates. The joint-perturbation holdout study tests a fixed algorithm beyond its development seed, but is not a statistical generalization theorem.

7.1. Model uncertainty and the economic margin for randomization

The certificates can be translated into a robustness statement without treating designed numerical primitives as estimated data. Fix a nominal model, initial law, and policy class. Suppose a nonempty set of alternative models has verified uniform bounds

$$|J_t^{p,M} - J_t^{p,0}| \leq a, \quad |V_t^M - V_t^0| \leq a, \quad |C_0^{p,M} - C_0^{p,0}| \leq b \quad (27)$$

for every common policy p , state, and date, and every M in the set. Superscript 0 denotes the nominal model. The error recursion in Section 4 supplies such bounds from uniform reward, cost, terminal, and kernel enclosures on a common state space; no reset aggregation is needed for this fixed-space perturbation statement. Uniformity over policies is essential. A fitted residual checked only under the proposed policy does not establish (27).

PROPOSITION 2—Portable operating and cost intervals: *Let $\varepsilon > 2a$. A nominal policy feasible at allowance $\varepsilon - 2a$ is feasible at allowance ε in every alternative model. If L_+ is a nominal lower bound at allowance $\varepsilon + 2a$ and U_- is the nominal cost of such a tightened feasible policy, then every model-specific common-policy optimum belongs to*

$$[\max\{0, L_+ - b\}, U_- + b].$$

The same interval encloses the optimum that minimizes worst-case implementation cost over policies feasible in all the alternative models. In the latter assertion the upper endpoint uses a single policy, not a different policy for each model.

PROOF: For the proposed policy, $J^{p,M} \geq J^{p,0} - a \geq V^0 - \varepsilon + a \geq V^M - \varepsilon$. Conversely a policy feasible in a model M satisfies $J^{p,0} \geq J^{p,M} - a \geq V^M - \varepsilon - a \geq V^0 - \varepsilon - 2a$. Its nominal cost is at least L_+ and its cost in M at least $L_+ - b$. Take infima for the model-specific statement. For the robust problem, the first implication supplies one policy for all models and the second applies to every robustly feasible policy in each model. Taking the worst-case cost preserves the bounds. *Q.E.D.*

A strict randomized separation also has an economically interpretable administration-cost allowance. Let U_R be the certified cost of a feasible common lottery and L_D a lower bound for every feasible deterministic policy, for the same operating constraint and initial law. Suppose operating payoffs and dynamics are unchanged by administering the lottery. A nonnegative extra administration charge at date t bounded by h_t increases its expected discounted cost by at most $H = \sum_{t < T} \beta^t h_t$. If

$$U_R + H < L_D, \tag{28}$$

then the lottery remains cheaper than every deterministic feasible policy even after this additional charge. This conclusion does not require administration at every visited state; charging the bound everywhere is conservative. Deterministic rules are assumed not to pay the randomization-specific charge.

For a common per-date ceiling h , the certified break-even threshold is

$$h_* = (L_D - U_R)_+ \bigg/ \sum_{t=0}^{T-1} \beta^t. \tag{29}$$

Every $h < h_*$ preserves the strict ranking. Equality at h_* yields no strict conclusion. For model uncertainty, use a deterministic nominal lower bound at $\varepsilon + 2a$ and a nominal lottery feasible at $\varepsilon - 2a$; then replace the numerator by $(L_{D,+} - U_{R,-} - 2b)_+$. The current development results report $a = b = 0$: they calculate economic margins under known designed primitives, not empirical uncertainty radii.

TABLE IV
CERTIFIED ALLOWANCE FOR A PER-DATE RANDOMIZATION ADMINISTRATION CHARGE

T	Installed rule	ε	Det. lower	Lottery upper	h_* lower bound
4	defer	0.01	4.93668	4.73011	0.055684
4	defer	0.05	4.93206	3.89012	0.280857
4	neural31001	0.01	0.11117	0.08928	0.005902
8	defer	0.01	10.81072	10.63794	0.025667
8	defer	0.05	10.75071	9.85440	0.133151
8	neural31001	0.01	0.46864	0.40232	0.009853
12	defer	0.01	15.61281	15.46691	0.015872
12	defer	0.05	15.47129	14.72413	0.081278
12	neural31001	0.01	3.49393	3.36015	0.014553

Uniform initial law, unchanged development economy. Endpoints and the threshold are rounded outward. A positive threshold means that every smaller constant per-date overhead preserves a strict cost advantage over every feasible deterministic policy. All 126 rule/horizon/tolerance/initial-law margins are retained in the machine-readable results, including zeros.

8. CONCLUSION

A local operating requirement becomes a global revision method only when one can control both continuation compatibility and the cost of restoring feasibility. The backward repair provides that control for positive tolerance and strict discounting. Combined with complete policy search, it produces a convergent certificate for the actual common randomized Markov objective. Combined with a verified reset approximation, it yields an explicit end-to-end error budget for an unrestricted measurable-policy problem. These results retain the optimization target rather than replacing it by a necessary-action or finite-price relaxation.

The computations show both the benefit and the burden of this construction. Whole-domain lotteries resolve a substantive deterministic-versus-randomized comparison on the original development cohort. Connected support bounds improve the same primary objective, and nonzero nonlinear lower bounds replace feasibility-only intervals. A matched direct global-programming comparator is often stronger per explored box. Longer horizons and four-state joint perturbations retain material gaps under fixed budgets. The stopping theorem explains how complete refinement can close them; the reported execution does not claim to have done so. The remaining numerical challenge is to exploit economic structure strongly enough that this complete certification becomes practical at larger state and horizon dimensions without changing its policy class or silently weakening its operating constraints.

REFERENCES

- Altman, E. (1999): *Constrained Markov Decision Processes*. Chapman and Hall/CRC. [2]
Brown, D. B., and J. E. Smith (2014): “Information Relaxations, Duality, and Convex Stochastic Dynamic Programs,” *Operations Research*, 62(6), 1394–1415. doi:10.1287/opre.2014.1322. [3]
Brown, D. B., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58(4, part 1), 785–801. doi:10.1287/opre.1090.0796. [3]
Brumm, J., and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85(5), 1575–1612. [3]
Ding, D., K. Zhang, T. Basar, and M. R. Jovanovic (2020): “Natural Policy Gradient Primal-Dual Method for Constrained Markov Decision Processes,” *Advances in Neural Information Processing Systems*, 33. [3]
Hettich, R., and K. O. Kortanek (1993): “Semi-Infinite Programming: Theory, Methods, and Applications,” *SIAM Review*, 35, 380–429. doi:10.1137/1035089. [3]
Huangfu, Q., and J. A. J. Hall (2018): “Parallelizing the Dual Revised Simplex Method,” *Mathematical Programming Computation*, 10, 119–142. doi:10.1007/s12532-017-0130-5. [3]

- Judd, K. L. (1998): *Numerical Methods in Economics*. Cambridge, MA: MIT Press. [3]
- Kushner, H. J., and P. G. Dupuis (2001): *Numerical Methods for Stochastic Control Problems in Continuous Time*, second edition. New York: Springer. [15]
- Laroche, R., P. Trichelair, and R. Tachet des Combes (2019): “Safe Policy Improvement with Baseline Bootstrapping,” *Proceedings of Machine Learning Research*, 97, 3652–3661. [3]
- McCormick, G. P. (1976): “Computability of Global Solutions to Factorable Nonconvex Programs: Part I—Convex Underestimating Problems,” *Mathematical Programming*, 10, 147–175. doi:10.1007/BF01580665. [3]
- Munos, R., and C. Szepesvári (2008): “Finite-Time Bounds for Fitted Value Iteration,” *Journal of Machine Learning Research*, 9(27), 815–857. [3]
- Ni, T., and M. Kamgarpour (2025): “A Safe Exploration Approach to Constrained Markov Decision Processes,” *Proceedings of Machine Learning Research*, 258, 3592–3600. [3]
- Rump, S. M. (2010): “Verification Methods: Rigorous Results Using Floating-Point Arithmetic,” *Acta Numerica*, 19, 287–449. [3]
- Rust, J. (1987): “Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher,” *Econometrica*, 55, 999–1033. doi:10.2307/1911259. [3]
- Saldi, N. (2018): “Finite-State Approximations to Discounted and Average Cost Constrained Markov Decision Processes,” arXiv:1807.02994. [2]
- Saldi, N., T. Linder, and S. Yüksel (2018): *Finite Approximations in Discrete-Time Stochastic Control: Quantized Models and Asymptotic Optimality*. Birkhäuser. doi:10.1007/978-3-319-79033-6. [2]
- Scherrer, B. (2013): “Performance Bounds for λ Policy Iteration and Application to the Game of Tetris,” *Journal of Machine Learning Research*, 14(36), 1181–1227. [3]

SUPPLEMENT TO “CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION”

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This supplement supplies proof details, exact model contracts, complete result tables, and the distinction between independent verification and inherited evidence. The complete R38 article and supplement are appended without modification. The main article is self-contained and does not require the archival annex.

1. OBJECTS, ASSUMPTIONS, AND VERIFICATION BOUNDARIES

The economic object throughout the new theory is one randomized Markov policy used at all states and dates. Each inequality $J_t^p(x) \geq V_t(x) - \varepsilon$ concerns that same policy. Its cost is integrated under a named initial law. None of the new convergence results treats a collection of Dirac-optimal policies as a solution under a dispersed initial law.

TABLE I: Assumptions and certificate scope

Result	Target and assumptions	Computation and check
Repair and modulus	Borel state space, finite actions, bounded rewards and costs, $0 < \beta < 1$, $\varepsilon > 0$; common randomized Markov policy; any initial law	Analytic proof. Borel measurability and the model bounds are assumptions, not facts inferred from a hash.
Witness-driven repair	Checked $L \leq V \leq U$, $U_T = g$, and positive residual margin $(1 - \beta)\varepsilon - d$	Analytic bracket plus 48 exact multi-action test cases. A failed margin check is not economic infeasibility.
Model perturbation	Uniform error bounds over all common policies and every alternative model; $\varepsilon > 2a$	One tightened policy is portable across models. Error radii are assumptions, not estimated confidence sets here.
Administration charge	Same operating model and initial law, feasible lottery upper, valid deterministic lower, bounded non-negative overhead	All 126 exact known-model margins retained. Strict superiority requires overhead strictly below its threshold.
Finite complete search	Finite states; same all-restart constraints and initial masses; compact policy product	Rational evaluation and verified box search. Full independent SymPy checks for 16 executed search trees.
Continuous transfer	Certified reset-kernel approximation in operator total variation; initial mixture identity; $\varepsilon > 2a_h$	Analytic end-to-end theorem. No claim that atomic benchmark kernels satisfy a vanishing reset error.

Result	Target and assumptions	Computation and check
Primary lotteries	Unchanged one-state, atomic affine-shock model; all-state operating feasibility; uniform, $2x$, or $2(1-x)$ initial density	Exact operating identities including isolated points; directed cost boxes; 42 independent lottery and local-bound checks.
Deterministic restrictions	Borel finite-action necessary correspondence; deterministic policies only	R38 proof objects and independent verification remain retained. The deterministic lower bound is not a randomized lower bound.
Nonlinear support	Two-state bilinear dynamics; uniform square; $\varepsilon = 1/2$ for paired upper policies	Directed vector arithmetic on all 20 meshes; independent scalar checks for two inherited objects and a new complete (32, 128) replay. Four $\varepsilon = 1/20$ rows are lower-only.
Retained local adaptive LP	A continuous one-step relaxation, not the common-policy global optimum	Prior exact whole-cell certificates retained; new connected lower recursion independently checked.
Stopped diffusion	Continuous actions, boundary settlement and diffusion approximation still require coverage	Archival derivation, not a solved numerical target or external validation.

2. DETAILS OF THE REPAIR AND COMPLETE-SEARCH ARGUMENTS

2.1. Policy perturbation and terminal rewards

Use total variation $\text{TV}(p, q) = \frac{1}{2} \sum_a |p_a - q_a|$ for action distributions. If an action-dependent scalar takes values in an interval of length D , replacing p by q changes its expectation by at most $D \text{TV}(p, q)$. With h stages remaining, a stage cost at distance s can be affected by the current decision and by s earlier transitions. Thus the cost perturbation constant is $K \sum_{s=0}^{h-1} (s+1)\beta^s$. Operating rewards contribute $2R \sum_{s=0}^{h-1} (s+1)\beta^s$, and the terminal reward contributes $2Gh\beta^h$. The maximum over all remaining horizons is necessary: $h\beta^h$ need not be increasing in h . Taking only the horizon- T terminal term would not provide a uniform all-restart bound. The main article uses this maximum explicitly.

2.2. Common-continuation repair and approximate minimizers

Let the original policy be feasible at allowance $\varepsilon + \delta$. At each backward step, all next-date repairs have already been fixed. The old current action probabilities evaluated against that repaired continuation give $q_p \geq V_t - \varepsilon - \delta$. The best current action gives $q_* \geq V_t - \beta\varepsilon$. If $q_p < V_t - \varepsilon$, the required mixture share is

$$\alpha = \frac{V_t - \varepsilon - q_p}{q_* - q_p} \leq \frac{\delta}{(1 - \beta)\varepsilon + \delta}.$$

Mixing occurs only on the Borel set with a positive numerator and denominator. The first maximizer in a fixed action ordering is Borel because the action set is finite. Operating value weakly

improves at every date and state. The same policy therefore satisfies all repaired constraints simultaneously. A coin selected once at time zero and remembered would not prove this result; the construction does not use such a coin.

For a general Borel model an optimizer need not exist. Apply the repair to a policy whose cost is within $\xi > 0$ of the relaxed infimum. The repaired cost is at most that relaxed infimum plus $\xi + D_C \rho_\varepsilon(\delta)$. Letting $\xi \downarrow 0$ proves the value modulus without assuming attainment. In a finite-state model, the product of policy simplexes is compact, the finite-horizon Bellman values are polynomials in its coordinates, and the feasible set is closed and nonempty. There the constrained minimum is attained. The pure-action finite-state problem is a finite enumeration; the randomized repair does not preserve deterministic policies and is not presented as a deterministic repair.

2.3. Why necessary contractions preserve completeness

Fix the already contracted next-date probability intervals. Their rectangular operating Bellman recursion has one maximizing Markov policy that attains its upper value at all states, by backward induction. The inequality used to contract a current interval is therefore necessary for every feasible policy remaining in the original box. Contracting a current coordinate leaves next-date values unchanged. Its retained endpoint set, when nonempty, still permits an operating value at least $V_t - \varepsilon$. Proceeding backward establishes this fact simultaneously at all dates of the contracted box. Capping an operating upper value by V_t does not reduce a true rectangular maximum, since an unconstrained operating optimum already dominates every box-restricted policy.

The implementation also computes lower operating enclosures and may discard a box when a lower enclosure exceeds its upper enclosure. That is an inconsistency test, not a failure of one sufficient candidate. Lower operating clipping at $V_t - \varepsilon$ uses a property of every feasible policy. Such additional necessary tests cannot remove a feasible policy. The separate cost lower recursion uses the retained probability rectangle without imposing incompatible, restart-specific continuation choices. Ordinary cost Bellman minimization attains this rectangular lower value at every state.

Let w be the largest retained interval width. The midpoint differs by at most w in total variation from the rectangular operating maximizer and the rectangular cost minimizer. Hence its operating deficit is at most $D_J w$, and its cost is at most the rectangular lower cost plus $D_C w$. Repair adds at most $D_C \rho_\varepsilon(D_J w)$. This proves the stated box error, including the effect of contractions.

For the finite stopping count, put $a = \beta^{T-1} w_*$. Any physical width exceeding w_* has weighted width exceeding or equal to a . Each weighted coordinate starts at at most one and, whenever selected, is at least halved; contractions can only reduce it further. Along any path, after at most $\lceil \log_2(1/a) \rceil$ halvings per coordinate, no weighted width can remain above a . Because the largest weighted coordinate is selected, no more than nT times that number of splits are needed before all physical widths are at most w_* . A positive-gap search never needs to refine a box whose repaired midpoint is already within the requested gap of its lower bound. There are finitely many possible nodes above this depth. The count can be enormous, and does not imply that the fixed-budget experiments exhaust it.

2.4. Rational policy lattices and a deterministic benchmark

For m actions, rounding down the first $m - 1$ probabilities and assigning the remainder to the last action gives a total-variation error at most $(m - 1)/M$. The relaxed feasibility test at

allowance $\varepsilon + D_J \theta_M$ retains a rounded version of every originally feasible policy. Its minimum cost minus $D_C \theta_M$ is a lower bound. Repair of its minimizing policy gives a feasible upper bound. The main theorem bounds the gap by $D_C[\theta_M + \rho_\varepsilon(D_J \theta_M)]$. Lattice levels need not be nested: taking the maximum of all lower endpoints and the minimum of all feasible upper endpoints yields nested certificates. The executed algorithm uses box search, rather than asserting that this exhaustive lattice was run.

3. CONTINUOUS-STATE TRANSFER WITHOUT A HIDDEN POLICY SIEVE

3.1. Averaging identity

In the reset model, after a transition into E_j , the conditional within-cell law is exactly μ_j , independently of the predecessor state and action. For an arbitrary Borel policy p , define $\bar{p}_t(a | i) = \int_{E_i} p_t(a | x) \mu_i(dx)$. At the terminal date the payoff is cellwise constant. If the averaged next-date values coincide with those of the finite model, integrating the reset Bellman equation over μ_i gives the finite Bellman equation with action probabilities $\bar{p}_t$. Backward induction proves the identity for operating values and costs. It uses the same averaged policy at every restart, not a separately optimized policy for each i .

Uniform primitive comparison bounds $|J^p - J_h^p|$ by a_t and $|C^p - C_h^p|$ by b_t for the very same policy. The comparison for optimal operating values is obtained directly from the contraction and monotonicity of finite-horizon Bellman operators. An original feasible policy, when averaged, therefore satisfies the finite constraints at allowance $\varepsilon + 2a_h$. A finite policy feasible at $\varepsilon - 2a_h$ can be lifted to the original model and satisfies its original constraints. The initial-mixture identity converts the integrated cost exactly in the reset model; b_h controls the remaining original-model error. Combining both maps with the repair modulus produces the global error budget in the article.

3.2. Verification obligations for an application

The theorem needs a bound for every bounded Borel test function, not merely weak convergence for smooth functions. A verified uniform density approximation can provide such a bound through an L_1 estimate. For a compact domain with a Lipschitz density in both source and destination coordinates, piecewise-constant density integration gives a computable linear mesh modulus, with cell volume and domain diameter included in the constant. Discontinuous installed rules require an aligned partition or a separately bounded cost error. The approximation must retain the declared initial-law mixture. These conditions are sufficient, not automatically satisfied by a chosen mesh.

The affine and bilinear benchmark kernels contain atoms. Replacing their images by uniform cell measures does not furnish a vanishing operator-total-variation error. Their direct whole-domain certificates do not depend on this transfer theorem. We retain their unresolved structural gaps and do not present a mesh sequence as a proof of convergence for those atomic models. The stopped diffusion would additionally need continuous-action coverage, boundary settlement, time discretization, and diffusion/quadrature error bounds before the same end-to-end conclusion could be claimed.

4. EXACT MODEL CONTRACTS AND PROSPECTIVE EVALUATION

4.1. Finite-state joint perturbations

The protocol commits seeds 391101–391108 after development seed 390001, with the case dimensions and horizons in Table II. The two-condition state rewards are $(1/5, 4/5)$. The

four-state rewards are $(1/5, 1/2, 1/2, 4/5)$, representing the average of two low/high indicators. Action $a \in \{0, 1\}$ has operating reward $s_i - ac$, terminal reward $s_i/2$, and revision cost $a(1 + s_i)$ against the installed action zero. The seed fixes $c \in \{11, \dots, 20\}/100$ and transition jitters; the complete generated rational matrices are inside every proof object.

For two states, the probability of the high-condition destination is

$$P^a(i, \text{high}) = \{22\mathbf{1}_{i=0} + 65\mathbf{1}_{i=1} + j_{ai} + 40a\}/120, \quad j_{ai} \in \{0, \dots, 8\}.$$

For four states, writing the source as $(i_x, i_y) \in \{0, 1\}^2$, the two destination indicators are conditionally independent given the source and action, with probabilities

$$p_x = (18 + 39i_x + 7i_xi_y + j_x + 37a)/120, \quad p_y = (20 + 36i_y + 5i_xi_y + j_y + 37a)/120,$$

where the seed gives $j_x, j_y \in \{0, \dots, 6\}$. Their product probabilities specify the four-state row. Odd seeds use uniform initial masses; even seeds use $\nu_i = (i + 1)/\sum_{j=1}^n j$. This is a designed joint-perturbation suite, not a calibrated economic panel or a random sample of environments.

The protocol at commit 13fd3ee0eec78db5fa6aed34e63819c5ed546209 fixes the two solver hashes, all eight cases, 4,096 nodes, 60 seconds per solver/case, and absolute stopping tolerance $1/1000$. No solver coefficient was changed after these eight environments were generated. A subsequent CI execution is a replay of the frozen experiment, not a second untouched holdout. Wall-clock stopping can change the explored node count across machines. The first local execution summary is retained separately; tables identify the published execution by its manifest.

TABLE II: Complete finite-state work and termination records

Seed	Method	Absolute gap	Nodes	Stop	MiB RSS	Seconds	Gap met
391101	I	0.000993	1057	gap	19.8	0.99	yes
391101	M	0.000981	241	gap	77.4	1.77	yes
391102	I	0.000000	1	budget	19.0	0.00	yes
391102	M	0.000000	1	budget	76.0	0.38	yes
391103	I	1.751049	4095	budget	32.6	17.19	no
391103	M	0.475442	4077	time_cap	115.7	60.41	no
391104	I	0.413280	4095	budget	36.2	16.32	no
391104	M	0.110602	3035	time_cap	117.2	60.40	no
391105	I	0.005913	4095	budget	30.2	10.82	no
391105	M	0.000999	2985	gap	97.8	35.99	yes
391106	I	0.000974	159	gap	19.0	0.47	yes
391106	M	0.000763	157	gap	77.8	2.60	yes
391107	I	3.184929	3815	time_cap	46.3	60.04	no
391107	M	1.649036	1529	time_cap	105.6	60.41	no
391108	I	1.750167	2353	time_cap	41.9	60.04	no
391108	M	0.783226	957	time_cap	102.1	60.43	no

I and M denote interval Bellman and verified McCormick search. The memory figure is process peak resident memory; it includes interpreter and imported libraries, not only the search tree. Seconds include model construction and the operating solve. The search cap is checked between proof-producing iterations, so a final iteration can exceed 60 seconds. Serialization

and independent checking are separate costs recorded in logs. A stopped run supplies a valid remaining interval; it is not a failed proof and is not labeled exact.

4.2. The nonlinear primitive contract

The state lies in $[0, 1]^2$, $\beta = 19/20$, and shock $z \in \{0, 1\}$ has probabilities $3/5, 2/5$. For each component, the transition is $(s \text{ own} + bxy + c + dz)/100$. The coefficient pairs for actions 0, 1, 2, 3 are

a	x component (s, b, c, d)	y component (s, b, c, d)
0	(65, 5, 0, 3)	(70, 3, 0, 2)
1	(15, 3, 70, 2)	(70, 3, 0, 2)
2	(65, 5, 0, 3)	(15, 4, 70, 2)
3	(10, 2, 80, 2)	(10, 2, 80, 2).

Operating costs are $(0, 9/50, 9/50, 8/25)$, reward is $(x + y)/2 - c_a$, and terminal reward is $(x + y)/4$. The installed action is 3 when both coordinates are below $1/4$; otherwise it is 1 when $x < 9/20$, 2 when $y < 9/20$, and 0 otherwise. Revision cost is $(1 + (x + y)/2)\mathbf{1}\{a \neq a^0(x, y)\}$. The independent checker separately partitions threshold-crossing boxes and verifies all transition ranges by rational corner geometry. This tests the paper-to-code model contract as well as arithmetic direction.

5. COMPLETE CONTINUOUS-STATE RESULTS

The tables below have identical all-restart constraints on both sides. Lower and upper decimals are rounded outward, and the gap column is rounded upward. Relative gaps are descriptive decimal percentages calculated from the exact rational interval. “Gain” means the lottery upper cost is strictly below the deterministic lower cost for that same initial density. It does not mean that the lottery is a globally optimal randomized rule. Zero upper cost is exact by nonnegativity; no positive randomized row is relabeled exact.

5.1. Uniform initial density

T	Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
4	defer	0.01	4.61741	4.73011	0.11269	2.38	yes
4	defer	0.05	3.66404	3.89012	0.22608	5.81	yes
4	neural31001	0.01	0.08510	0.08928	0.00418	4.68	yes
4	neural31001	0.05	0.02622	0.03016	0.00394	13.04	yes
4	neural31002	0.01	0.12047	0.12517	0.00469	3.75	yes
4	neural31002	0.05	0.04796	0.05403	0.00607	11.22	yes
4	neural31003	0.01	0.09478	0.09909	0.00430	4.34	yes
4	neural31003	0.05	0.03178	0.03723	0.00544	14.61	yes
4	occupancy_stress	0.01	3.90691	4.02072	0.11381	2.83	yes
4	occupancy_stress	0.05	3.02022	3.24805	0.22783	7.01	yes
4	spline129	0.01	0.00000	0.00000	0.00000	0.00	no
4	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
4	spline33	0.01	0.00000	0.00000	0.00000	0.00	no

T	Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
4	spline33	0.05	0.00000	0.00000	0.00000	0.00	no
8	defer	0.01	10.50355	10.63794	0.13439	1.26	yes
8	defer	0.05	9.44911	9.85440	0.40528	4.11	yes
8	neural31001	0.01	0.34344	0.40232	0.05887	14.63	yes
8	neural31001	0.05	0.10408	0.14046	0.03638	25.89	no
8	neural31002	0.01	0.22548	0.27772	0.05223	18.81	yes
8	neural31002	0.05	0.04029	0.05952	0.01922	32.29	no
8	neural31003	0.01	0.55282	0.65391	0.10108	15.46	yes
8	neural31003	0.05	0.08634	0.10044	0.01410	14.03	yes
8	occupancy_stress	0.01	9.17362	9.36573	0.19210	2.05	yes
8	occupancy_stress	0.05	7.98380	8.56599	0.58218	6.80	yes
8	spline129	0.01	0.00000	0.00000	0.00000	0.00	no
8	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
8	spline33	0.01	0.00000	0.00000	0.00000	0.00	no
8	spline33	0.05	0.00000	0.00000	0.00000	0.00	no
12	defer	0.01	15.31722	15.46691	0.14969	0.97	yes
12	defer	0.05	14.14203	14.72413	0.58209	3.95	yes
12	neural31001	0.01	2.77095	3.36015	0.58920	17.53	yes
12	neural31001	0.05	1.58477	2.02354	0.43876	21.68	yes
12	neural31002	0.01	2.05173	2.38505	0.33332	13.98	yes
12	neural31002	0.05	1.19573	1.45515	0.25942	17.83	yes
12	neural31003	0.01	0.77248	1.01028	0.23780	23.54	yes
12	neural31003	0.05	0.09346	0.14211	0.04865	34.23	no
12	occupancy_stress	0.01	13.42003	13.65506	0.23503	1.72	yes
12	occupancy_stress	0.05	12.15841	12.87752	0.71911	5.58	yes
12	spline129	0.01	0.00000	0.00000	0.00000	0.00	no
12	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
12	spline33	0.01	0.00000	0.00000	0.00000	0.00	no
12	spline33	0.05	0.00000	0.00000	0.00000	0.00	no

5.2. Initial density $2x$

T	Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
4	defer	0.01	4.80430	4.91856	0.11426	2.32	yes
4	defer	0.05	3.85831	4.08355	0.22523	5.52	yes
4	neural31001	0.01	0.12619	0.13251	0.00632	4.77	yes
4	neural31001	0.05	0.03825	0.04405	0.00580	13.16	yes
4	neural31002	0.01	0.17588	0.18292	0.00704	3.85	yes
4	neural31002	0.05	0.06877	0.07759	0.00882	11.35	yes
4	neural31003	0.01	0.13995	0.14645	0.00650	4.43	yes
4	neural31003	0.05	0.04616	0.05411	0.00795	14.68	yes
4	occupancy_stress	0.01	3.63981	3.75184	0.11203	2.99	yes
4	occupancy_stress	0.05	2.80900	3.02583	0.21682	7.17	yes
4	spline129	0.01	0.00000	0.00000	0.00000	0.00	no

T	Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
4	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
4	spline33	0.01	0.00000	0.00000	0.00000	0.00	no
4	spline33	0.05	0.00000	0.00000	0.00000	0.00	no
8	defer	0.01	10.69052	10.82469	0.13417	1.24	yes
8	defer	0.05	9.68566	10.04214	0.35647	3.55	yes
8	neural31001	0.01	0.30165	0.34589	0.04424	12.79	yes
8	neural31001	0.05	0.11746	0.14930	0.03183	21.32	no
8	neural31002	0.01	0.11812	0.15183	0.03371	22.20	yes
8	neural31002	0.05	0.01393	0.02361	0.00967	40.94	no
8	neural31003	0.01	0.59210	0.68149	0.08939	13.12	yes
8	neural31003	0.05	0.10049	0.11544	0.01494	12.94	yes
8	occupancy_stress	0.01	8.90083	9.08695	0.18611	2.05	yes
8	occupancy_stress	0.05	7.75429	8.29459	0.54029	6.51	yes
8	spline129	0.01	0.00000	0.00000	0.00000	0.00	no
8	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
8	spline33	0.01	0.00000	0.00000	0.00000	0.00	no
8	spline33	0.05	0.00000	0.00000	0.00000	0.00	no
12	defer	0.01	15.50452	15.65425	0.14972	0.96	yes
12	defer	0.05	14.36785	14.91248	0.54462	3.65	yes
12	neural31001	0.01	3.00897	3.57101	0.56203	15.74	yes
12	neural31001	0.05	1.75968	2.30639	0.54670	23.70	yes
12	neural31002	0.01	2.29566	2.59770	0.30203	11.63	yes
12	neural31002	0.05	1.37651	1.66452	0.28800	17.30	yes
12	neural31003	0.01	0.70594	0.93114	0.22519	24.18	yes
12	neural31003	0.05	0.09731	0.14941	0.05209	34.86	yes
12	occupancy_stress	0.01	13.14135	13.37437	0.23301	1.74	yes
12	occupancy_stress	0.05	11.89664	12.60188	0.70523	5.60	yes
12	spline129	0.01	0.00000	0.00000	0.00000	0.00	no
12	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
12	spline33	0.01	0.00000	0.00000	0.00000	0.00	no
12	spline33	0.05	0.00000	0.00000	0.00000	0.00	no

5.3. Initial density $2(1 - x)$

T	Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
4	defer	0.01	4.43053	4.54166	0.11113	2.45	yes
4	defer	0.05	3.46976	3.69670	0.22693	6.14	yes
4	neural31001	0.01	0.04401	0.04605	0.00204	4.42	yes
4	neural31001	0.05	0.01419	0.01627	0.00207	12.72	yes
4	neural31002	0.01	0.06507	0.06742	0.00235	3.47	yes
4	neural31002	0.05	0.02714	0.03047	0.00332	10.88	yes
4	neural31003	0.01	0.04961	0.05173	0.00211	4.07	yes
4	neural31003	0.05	0.01741	0.02036	0.00294	14.44	yes
4	occupancy_stress	0.01	4.17401	4.28960	0.11559	2.69	yes

T	Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
4	occupancy_stress	0.05	3.23143	3.47027	0.23883	6.88	yes
4	spline129	0.01	0.00000	0.00000	0.00000	0.00	no
4	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
4	spline33	0.01	0.00000	0.00000	0.00000	0.00	no
4	spline33	0.05	0.00000	0.00000	0.00000	0.00	no
8	defer	0.01	10.31657	10.45120	0.13462	1.29	yes
8	defer	0.05	9.21257	9.66666	0.45409	4.70	yes
8	neural31001	0.01	0.38523	0.45874	0.07350	16.02	yes
8	neural31001	0.05	0.09070	0.13163	0.04092	31.08	no
8	neural31002	0.01	0.33285	0.40361	0.07075	17.53	yes
8	neural31002	0.05	0.06082	0.09542	0.03460	36.26	no
8	neural31003	0.01	0.51355	0.62632	0.11277	18.00	yes
8	neural31003	0.05	0.07220	0.08545	0.01325	15.50	yes
8	occupancy_stress	0.01	9.44731	9.64451	0.19720	2.04	yes
8	occupancy_stress	0.05	8.21366	8.83739	0.62372	7.06	yes
8	spline129	0.01	0.00000	0.00000	0.00000	0.00	no
8	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
8	spline33	0.01	0.00000	0.00000	0.00000	0.00	no
8	spline33	0.05	0.00000	0.00000	0.00000	0.00	no
12	defer	0.01	15.12991	15.27957	0.14965	0.98	yes
12	defer	0.05	13.91621	14.53578	0.61956	4.26	yes
12	neural31001	0.01	2.53293	3.14930	0.61637	19.57	yes
12	neural31001	0.05	1.40986	1.74069	0.33082	19.00	yes
12	neural31002	0.01	1.80779	2.17240	0.36460	16.78	yes
12	neural31002	0.05	1.01494	1.24578	0.23084	18.53	yes
12	neural31003	0.01	0.83901	1.08942	0.25041	22.98	yes
12	neural31003	0.05	0.08961	0.13482	0.04521	33.53	no
12	occupancy_stress	0.01	13.69954	13.93576	0.23621	1.69	yes
12	occupancy_stress	0.05	12.42017	13.15316	0.73298	5.57	yes
12	spline129	0.01	0.00000	0.00000	0.00000	0.00	no
12	spline129	0.05	0.00000	0.00000	0.00000	0.00	no
12	spline33	0.01	0.00000	0.00000	0.00000	0.00	no
12	spline33	0.05	0.00000	0.00000	0.00000	0.00	no

6. DETERMINISTIC GAPS AND THE EXCEPTIONAL RESTART LOCUS

The inherited deterministic headline decomposes into 12 zero-cost and 12 positive-cost equalities. Of the positive-cost equalities, ten occur at horizon 4, two at horizon 8, and none at horizon 12. The table reports every inherited absolute and relative interval and the outer selector's maximum operating deficit. A deficit above ε is an actual violation established by the exact one-dimensional representation; it is distinct from a failed sufficient nonlinear box certificate.

The installed policy has zero *revision* cost by definition. Therefore a revision gap divided by the installed policy's revision cost is undefined for every row. We do not report that ratio as zero, and do not substitute its operating reward as a denominator with different units. The valid cost normalization is gap divided by a positive feasible upper revision cost; a zero upper cost is separately labeled exact.

T	Rule	ε	Det. gap	Gap/upper (%)	Deficit	Excess
4	defer	0.01	0.000000	0.00	0.01000	0.00000
4	defer	0.05	0.000000	0.00	0.05000	0.00000
4	neural31001	0.01	0.000000	0.00	0.01000	0.00000
4	neural31001	0.05	0.000000	0.00	0.05000	0.00000
4	neural31002	0.01	0.000000	0.00	0.01000	0.00000
4	neural31002	0.05	0.000000	0.00	0.05000	0.00000
4	neural31003	0.01	0.000000	0.00	0.01000	0.00000
4	neural31003	0.05	0.000000	0.00	0.05000	0.00000
4	occupancy_stress	0.01	0.000000	0.00	0.01000	0.00000
4	occupancy_stress	0.05	0.000000	0.00	0.05000	0.00000
4	spline129	0.01	0.000000	–	0.00001	0.00000
4	spline129	0.05	0.000000	–	0.00001	0.00000
4	spline33	0.01	0.000000	–	0.00001	0.00000
4	spline33	0.05	0.000000	–	0.00001	0.00000
8	defer	0.01	0.001622	0.01	0.01170	0.00170
8	defer	0.05	0.019446	0.18	0.09072	0.04072
8	neural31001	0.01	0.010990	2.29	0.01170	0.00170
8	neural31001	0.05	0.103474	45.39	0.09072	0.04072
8	neural31002	0.01	0.010990	3.16	0.01170	0.00170
8	neural31002	0.05	0.069936	61.21	0.07754	0.02754
8	neural31003	0.01	0.006434	0.74	0.01170	0.00170
8	neural31003	0.05	0.000000	0.00	0.07693	0.02693
8	occupancy_stress	0.01	0.000000	0.00	0.01000	0.00000
8	occupancy_stress	0.05	0.002381	0.03	0.05763	0.00763
8	spline129	0.01	0.000000	–	0.00042	0.00000
8	spline129	0.05	0.000000	–	0.00042	0.00000
8	spline33	0.01	0.000000	–	0.00576	0.00000
8	spline33	0.05	0.000000	–	0.00576	0.00000
12	defer	0.01	0.001359	0.01	0.01170	0.00170
12	defer	0.05	0.066102	0.43	0.12738	0.07738
12	neural31001	0.01	0.014120	0.40	0.01170	0.00170
12	neural31001	0.05	0.177733	6.89	0.09072	0.04072
12	neural31002	0.01	0.009537	0.37	0.01170	0.00170
12	neural31002	0.05	0.075788	4.12	0.07897	0.02897
12	neural31003	0.01	0.006662	0.54	0.01170	0.00170
12	neural31003	0.05	0.248472	63.88	0.11215	0.06215
12	occupancy_stress	0.01	0.003444	0.03	0.01194	0.00194
12	occupancy_stress	0.05	0.060221	0.45	0.06919	0.01919
12	spline129	0.01	0.000000	–	0.00042	0.00000
12	spline129	0.05	0.000000	–	0.00042	0.00000
12	spline33	0.01	0.000000	–	0.00576	0.00000
12	spline33	0.05	0.000000	–	0.00576	0.00000

6.1. Exact disagreement for horizon 8, neural31003, allowance 1/20

A direct check reveals a stronger initial equality than the earlier verbal description suggested: $C_0 = B_0$ as represented functions, including isolated points. The failure of the all-restart equality

flag comes from dates 1, 2, and 3 below. These are not merely null-set changes to the initial cost function. They are distinct restarted problems. The nonnegative Bellman cost-gap decomposition and initial equality imply zero expected later gap under the selected policy and the original uniform law; positive-gap intervals therefore have zero probability along that policy from this initial law. They remain relevant when the controller is restarted at their states. The following fractions come directly from the independent piecewise-affine parser. No decimal equality test was used.

6.1.0.1. Restart date 1. Open interval with endpoints 0 and $7195348168316/157454650804455$. The difference has slope $55233/2000000$ and intercept $289161/500000$. At 0, the separately represented difference is $289161/500000$.

6.1.0.2. Restart date 2. Open interval with endpoints $11274755890307953/1459200000000000$ and $4598974049913359/5910038365284645$. The difference has slope $1031/2000$ and intercept $373/1000$.

Open interval with endpoints $36597632066686/46374770160891$ and $4487129063774/5675886884445$. The difference has slope $3061039/4000000$ and intercept $1229987/2000000$.

Open interval with endpoints $4487129063774/5675886884445$ and $633416277301978/787273254022275$. The difference has slope $969/2000$ and intercept $627/1000$.

At $11274755890307953/1459200000000000$, the separately represented difference is $22509905322907499543/2918400000000000000$.

At $36597632066686/46374770160891$, the separately represented difference is $56526876978886030897/46374770160891000000$.

At $4487129063774/5675886884445$, the separately represented difference is $2958098610911/2928734202500$.

6.1.0.3. Restart date 3. Open interval with endpoints $1288720115486/1669378495425$ and $181509165032032/231550957065375$. The difference has slope 1 and intercept 1.

At $1288720115486/1669378495425$, the separately represented difference is $2958098610911/1669378495425$.

7. NONLINEAR BOUNDS AND WORK

The support-only lower endpoint is shown beside its connected restart improvement. All upper policies are the same retained feasible candidates; thus the improvement is not confounded with a larger proposal portfolio. A missing upper endpoint means that no upper candidate at that mesh was certified by the inherited sufficient test. It does not imply true infeasibility. No comparison of unmatched success fractions is used.

T	N	Support lower	Restart lower	Upper	Gap/upper (%)	Array MiB
4	16	0.0000	0.0000	2.4975	100.00	0.1
4	32	0.0005	0.0005	2.1219	99.97	0.6
4	64	0.0118	0.0118	1.7191	99.31	2.3
4	128	0.0332	0.0334	1.3503	97.53	9.4
4	256	0.0554	0.0556	0.9992	94.43	37.5

T	N	Support lower	Restart lower	Upper	Gap/upper (%)	Array MiB
8	16	0.0000	0.0000	–	–	0.3
8	32	0.2521	0.2630	5.1263	94.87	1.1
8	64	0.9214	0.9438	3.9348	76.01	4.2
8	128	1.2956	1.3108	3.6152	63.74	16.9
8	256	1.4773	1.4895	3.0418	51.03	67.5
16	16	0.0168	0.0169	–	–	0.5
16	32	1.7800	1.8264	–	–	2.0
16	64	3.1646	3.1898	8.7556	63.57	8.0
16	128	4.1609	4.1908	7.6644	45.32	31.9
16	256	4.6932	4.7174	7.3118	35.48	127.5
32	16	0.4443	0.5535	–	–	1.0
32	32	3.4755	3.5226	–	–	3.9
32	64	6.4932	6.5389	16.7884	61.05	15.5
32	128	8.0961	8.1264	12.5226	35.11	61.9
32	256	8.8476	8.8718	11.8492	25.13	247.5

The array payload excludes temporary continuation ranges, the interpreter, and compression buffers; peak process memory is also saved. On a threshold-crossing box, the cost upper enclosure charges the largest possible realized departure cost, while the lower enclosure can charge zero if the installed action is possible there. This conservatism concerns enclosure of the implemented policy, not a claim that the policy changes at every point of that box. The nonzero supports address the earlier lower-bound weakness without removing this upper-enclosure conservatism.

7.1. Operating allowance 1/20

These four rows test a tolerance comparable to the primary development cohort. They are explicitly lower-bound-only diagnostics. A feasible policy at allowance 1/2 cannot be used as an upper policy at allowance 1/20.

T	N	ε	Lower	Witness width	Status
4	64	0.05	1.55879	0.10794	uncertified upper
8	128	0.05	4.24475	0.08548	uncertified upper
16	128	0.05	7.76989	0.13408	uncertified upper
32	256	0.05	12.45297	0.10222	uncertified upper

8. LOCAL-LP ERROR VERSUS WORK AND DEGENERACIES

The inherited adaptive calculation is retained in full. Its finest tolerances required substantially more cells and work. The table and two generated figures display all 12 executions, not only favorable rows. They concern a continuous local relaxation with zero global support floor, and are not silently combined with the new nonzero-floor global recursion as if their errors were additive.

Witness	ε	Local tolerance	Cells	Local interval width	Seconds
constructed	0.01	1/100	3159	0.018550	11.36
constructed	0.01	1/200	6275	0.009275	32.39
constructed	0.01	1/400	12529	0.004638	106.30
exact	0.01	1/100	3156	0.018550	8.91
exact	0.01	1/200	6272	0.009275	25.40
exact	0.01	1/400	12526	0.004638	82.27
constructed	0.05	1/100	171	0.018550	0.51
constructed	0.05	1/200	309	0.009275	1.03
constructed	0.05	1/400	588	0.004638	2.35
exact	0.05	1/100	168	0.018550	0.40
exact	0.05	1/200	306	0.009275	0.84
exact	0.05	1/400	585	0.004638	1.96

The new local calculation uses a fixed represented partition, not an adaptive nondegeneracy termination assumption. For one operating inequality, it enumerates feasible pure actions and two-action mixtures straddling the boundary. When a pure action lies exactly on the boundary it is already included. When two operating values coincide, no division by their difference is performed; either their face is infeasible or its linear cost minimum is attained at an included pure vertex. Thus rational coincidences do not create an unreported termination assumption for the new calculation. The old logs did not instrument an occurrence count for every degeneracy in their adaptive proof construction. We do not infer such a count from successful termination. Those old claims remain subject to their explicitly conditional theorem.

9. INDEPENDENT CHECKS, PUBLICATION CHECKS, AND PRESERVATION

The 16 finite proof trees are checked using SymPy rationals without importing either solver, the repair constructor, or the LP package. The verifier reconstructs the economic stage contract, common-policy Bellman equalities, operating feasibility, box coverage, contractions, all accepted nonnegative multiplier residuals, and the minimum over terminal leaves. Nine distinct corruptions of policy class, probabilities, terminal value, initial law, revision cost, lower endpoint, multiplier sign, or root coverage are rejected. These mutation tests supplement rather than prove the verifier's all-input correctness.

The 42 continuous-state lottery objects and their 42 connected local lower objects receive independent operating, cost, support, and vertex checks. The separate randomized and deterministic inherited endpoints retain their provenance. Most final lower endpoints are either the newly checked local integral or the newly checked exact support integral. A retained R34 randomized endpoint may dominate those at a particular row; such an inherited component is not converted into a fresh R39 independent proof merely by taking a maximum. The publication audit records the source components. Likewise deterministic strict separation uses the retained R38 deterministic certificates, whose own independent checks remain part of the archive. The independent label for new lotteries and local bounds does not expand automatically to all historical components.

The two nonlinear independent checks cover the whole $(T, N) = (8, 64)$ and $(32, 64)$ objects, including policy cost and operating feasibility. They use rational corner geometry and scalar arbitrary-precision integer recursions, rather than vectorized construction arithmetic. Other nonlinear meshes, except the new R40 $(32, 128)$ replay, have directed construction and structural checks, not the same independent status. These labels are distinct from hash identity, repeated execution, and publication-integrity checks.

A seeded adversarial test separately generates 200 discontinuous piecewise-affine pairs, tests their exact lower envelopes and positive affine pullbacks against independent SymPy evaluation at knots and open-cell representatives, and rejects 400 malformed partitions. The retained report states its seed and comparison count. This tests isolated values, threshold crossings, and discontinuities beyond the nine hand-selected finite-tree mutations. It is not a probability estimate of false acceptance.

10. ADDITIONAL WITNESS, ROBUSTNESS, AND ARITHMETIC AUDITS

The main article proves a witness-driven feasibility repair and a lattice error budget that requires checked operating witnesses rather than an exact operating optimum. Its repair slack is $(1 - \beta)\varepsilon - d > 0$, where d bounds the upper witness’s Bellman residual. Approximation of the operating value and approximation of the policy remain separately visible. The new fixed-space perturbation proposition preserves one policy across every model in its uncertainty set. Break-even administration charges are derived for all 126 primary initial-law configurations, including zero margins.

The new rational repair constructor does not compute or accept the exact optimum. In 48 designed test models, a separate exact operating oracle constructs TEST witnesses; it is not input to the constructor. Signed stage and terminal rewards, two or three actions, two or three states, and horizons 2 through 5 are covered. SymPy matrix evaluation checks every operating and cost value against the Fraction constructor. All 48 deliberately nonpositive witness-slack cases are rejected as uncertified, not declared infeasible. These tests supplement the analytic theorem.

The seeded piecewise-affine audit checks 200 discontinuous pairs, including separately represented knot values, with 5756 independent value comparisons and 400 malformed-partition rejections. An additional 120 local programs are checked by an independently assembled SymPy active-basis calculation; 59 contain coincident operating values and 64 have a pure action exactly on the operating boundary. These are measured degeneracy counts for the new audit, not retrospectively invented counts for the historical adaptive solver.

R40 replays all 16 finite proof trees, three declared full primary lottery/local objects, and the complete nonlinear $(T, N) = (32, 128)$ object using the frozen independent verifiers. The latter extends prior scalar verification to a finer mesh. The full inherited 42-primary-object verifier coverage is retained by exact hash linkage; the three new primary replays are not represented as 42 fresh executions. None of these tests is a statistical false-acceptance guarantee.

10.1. *Preserved scientific content*

The appended R38 article and supplement contain the complete deterministic proof, fixed-restart history-conditioned frontier, two-period randomized example, adaptive local-LP theorem, economic sensitivity analysis, nonlinear feasibility construction, and killed-operator/stopped-control derivations. They remain historical documents with their original claims and limitations; the R39 article and this supplement specify the current conclusions. The complete older 349-page `HISTORY_R38.pdf`, prior reviews, failed runs, and source hashes remain unchanged on the branch. No historical scientific content is deleted. Preservation is not presented as additional evidence that an unresolved economic optimum has been solved.

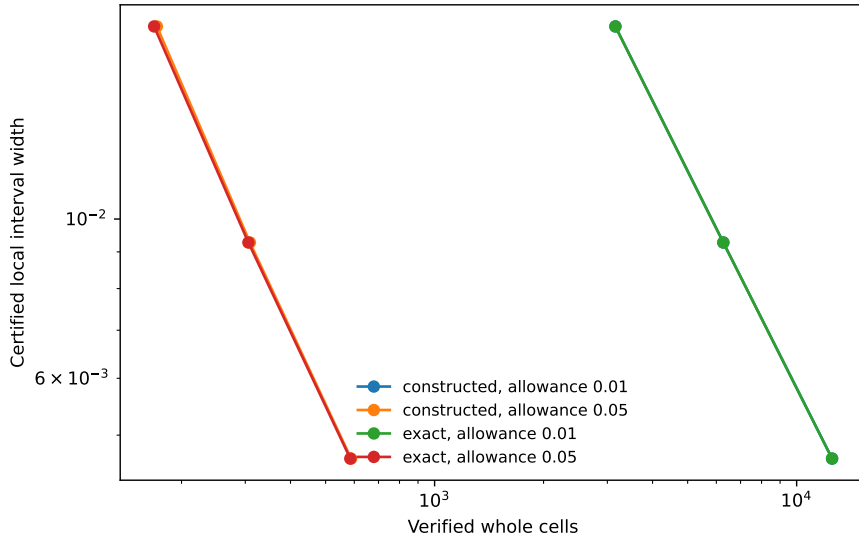


FIGURE 1.—Certified local interval width against whole cells. Every inherited local execution is displayed; the connected R39 global lower bounds solve a different recursive lower problem.

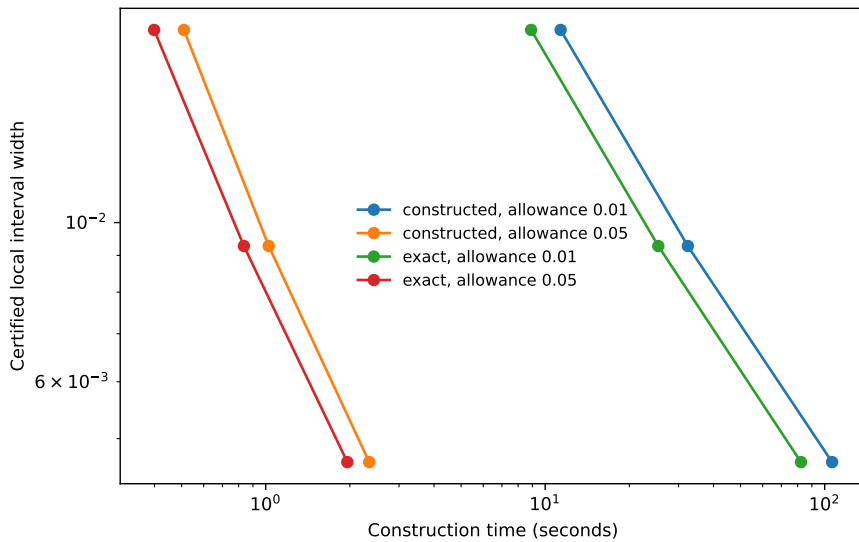


FIGURE 2.—Certified local interval width against construction seconds. The steep work increase is retained rather than hidden by an aggregate success count.

11. VERBATIM PRESERVED R38 ARTICLE

The following pages are the original article, not a newly edited substitute. Its initial cost-equality discussion is supplemented by the exact restart diagnostic above.

We study the least costly revision of an installed policy subject to operating guarantees at every state and restart. Two-sided Bellman witnesses generate safe outer restrictions for deterministic policies and cost lower bounds that can be matched by independently feasible deployments. In a continuous-state maintenance economy, exact arithmetic certifies 24 initial-distribution deterministic optima among 42 configurations, including 12 with positive implementation cost. An exact continuation-frontier recursion and a solved two-period randomized Markov example separate deterministic from randomized objectives. For randomized lower bounds, adaptive primal–dual cell certificates replace a finite local multiplier grid and provide a quantitative approximation guarantee. A nonlinear two-state economy tests directed enclosures beyond piecewise-affine closure, while 66 sensitivity configurations measure endogenous intervention exposure. Separate rational verification, unsuccessful runs, full computational costs, and nonzero residual intervals accompany the results. A killed-operator comparison connects the method to stopped economic control.

KEYWORDS: Dynamic programming, Policy revision, Verified computation, Constrained Markov control, Neural proposals.

1. INTRODUCTION

An installed decision rule can be inexpensive to operate and expensive to replace. A new rule to preserve operating performance, but its implementation cost depends on which decisions change and on the states the revised rule subsequently visits. Maintenance, organizational procedures, and deployed decision systems share this distinction. A rule that minimizes today’s number of changes need not minimize the discounted cost of future changes. The economic problem is therefore a constrained control problem, not a pointwise classification problem.

This paper constructs certificates for that problem. Its operating requirement is uniform over revision dates and states. Its implementation objective is an expectation under a specified initial distribution and the deployed policy’s own transition law. These are different scopes: a uniform performance constraint does not turn an initial-distribution optimum into a cost optimum from any restart. We maintain that distinction in the theory and in every numerical declaration of fitness.

The first contribution is a deterministic outer-class construction using two-sided operating witnesses. A feasible deterministic action must be compatible with the performance target even when followed by optimal operating continuation. Replacing that unavailable continuation by a certified upper witness, and optimal current value by a lower witness, preserves the necessity of the test. Minimizing implementation cost over the resulting outer class gives a valid lower bound. An independently evaluated feasible deployment gives an upper bound. Equality solves the constrained implementation objective, not merely an inner approximation. Exact operating dynamic programming supplies a reference specialization, and its full cost is charged.

The second contribution is a policy-class-specific constrained reference calculation. A continuation-frontier recursion solves finite-shock, finite-horizon problems from a fixed restart when continuation may depend on history. Its construction clips operating feasibility at each revision history, not only at the initial node. A two-period version of the maintenance economy admits an analytic randomized Markov solution for the uniform initial distribution. The

deterministic sufficiency is false even in the basic economic environment. A correct positive
 deterministic result cannot be extended to randomized policies by terminology alone.
 Our third contribution replaces a finite local multiplier grid by a statewise exact linear program
 and a single-cell approximation certificate. The exact primal frontier can be rational rather than
 affine. We bound it by a feasible primal expression and a fixed-multiplier dual function
 over each cell, using rational quadratic sign tests. Directed dyadic encoding controls
 growth without sacrificing the lower-bound direction. The resulting error bound
 relates to the continuous local restart relaxation. It does not assert global strong duality for
 the original constrained control problem. Separate experiments measure witness error, global
 refinement, local multiplier error, and feasible upper-search improvement.
 These constructions have material numerical consequences. Of 42 unchanged primary con-
 ditions, 24 have exactly matching deterministic lower and upper initial costs; 12 of these
 are strictly positive. All 14 horizon-four cases close. Of the 24 integrated equalities, 23
 are equalities of the complete cost-to-go functions. The remaining 18 deterministic cases
 explicitly reported nonzero intervals. The largest interval is below 0.249 in normalized
 cost units. No operating tolerance is relaxed to obtain these statements.
 Further studies examine the mechanism and the representation. Across 66 designed
 sensitivity tests, dynamic revision is strictly less costly than pointwise revision in 22
 cases and has exactly equal cost in 44. A separate bilinear two-state economy uses
 integer box enclosures, horizons through 32, and meshes through 256 per dimension.
 It produces unsuccessful certificates and an information-matched classical grid scalarization
 error. These are tests of a known-model numerical method, not empirical calibration or
 evidence of dimension-free scalability.

1.1. *Relation to existing methods*

In comparison, constrained backward induction, convexification, and weak duality are
 standard tools (Altman, 1999, Judd, 1998). We do not rename them as new principles. Our
 contribution is the constructive combination of operating witnesses, policy-class-correct outer
 bounds, a finite verifiable optimality test, and a numerical implementation that closes positive
 cost cases. The frontier computation provides a direct constrained reference for
 the history-conditioned objective, rather than comparing only with unconstrained operating
 costs.
 Approximate dynamic programming studies approximation and sampling errors under spec-
 ified information structures (Munos and Szepesvári, 2008, Scherrer, 2013). Safe policy im-
 plementation can restrict changes in response to uncertainty about a transition model (Laroche,
 et al., and Tachet des Combes, 2019). Here the model is known, approximation proposes a
 rule, a witness, and a separate deterministic verification layer checks the resulting rule on the
 state domain. Neural and spline proposals are retained on identical terms. We establish
 a neural speed advantage nor superiority of an installed neural rule merely because a
 deployment is feasible.
 Formal economics exploits economic and approximation structure (Judd, 1998, Brumm
 and Heiddegger, 2017). Information-relaxation methods provide a different route to bounds
 (Smith, and Sun, 2010, Brown and Smith, 2014). Both approaches must charge bound
 violation. Our one-dimensional reference calculation exploits exact affine structure; the
 multi-dimensional experiment replaces that structure with directed box arithmetic. Verified com-
 putation concerns the analytic reduction as well as arithmetic error (Rump, 2010). Accordingly,
 the arithmetic verification is specified by its actual code boundary, rather than inferred
 from the existence of a second execution path.

At $t = 0, \dots, T$, let X be a Borel state space, and let A be a finite action set. Rewards $r_t(x, a)$, terminal payoff $g(x)$, and implementation costs $k_t(x, a)$ are bounded and measurable, $k_t \geq 0$. The discount factor is $0 < \beta \leq 1$. A known transition kernel P_t^a defines

$$T_t^a v = r_t(\cdot, a) + \beta P_t^a v, \quad T_t v = \max_{a \in A} T_t^a v, \quad V_T = g, \quad V_t = T_t V_{t+1}. \quad (1)$$

Feasible selectors exist because the action set is finite. For a deterministic Markov policy π , the $J_T^\pi = g$, $C_T^\pi = 0$, and

$$J_t^\pi = T_t^{\pi_t} J_{t+1}^\pi, \quad C_t^\pi = k_t(\cdot, \pi_t) + \beta P_t^{\pi_t} C_{t+1}^\pi. \quad (2)$$

Given an installed rule π^0 and a nonnegative revision weight w_t , our principal specification is $r_t(x, a) = w_t(x) \mathbf{1}\{a \neq \pi_t^0(x)\}$. For a fixed initial distribution ν and a tolerance $\varepsilon \geq 0$, write

$$K_D(\nu, \varepsilon) = \inf_{\pi \in \mathcal{F}_D(\varepsilon)} \int C_0^\pi d\nu, \quad \mathcal{F}_D(\varepsilon) = \{\pi : V_t(x) - J_t^\pi(x) \leq \varepsilon \text{ for every } (t, x)\}. \quad (3)$$

Subscript D denotes deterministic Markov policies. A randomized Markov kernel $p_t(\cdot | x)$ averages the action-specific equations in (2); replacing the feasible class gives K_R . An implementation intervention is charged for the action actually drawn, so its conditional expected cost is $\sum_a p_t(a | x) k_t(x, a)$.

History dependence requires more notation. Let $h_t = (x_0, a_0, x_1, \dots, a_{t-1}, x_t)$ be an admissible decision history, augmented by any remembered past randomization. Continuation rewards $r_t(h_t)$ and costs $C_t^\pi(h_t)$ are defined before the date- t action draw. The history-conditioned feasible class requires

$$V_t(x_t) - J_t^\pi(h_t) \leq \varepsilon \quad \text{at every admissible decision history,} \quad (4)$$

including continuation kernels specified at off-path histories. We do not impose the same equality after conditioning on the current action draw but before receiving its payoff. Such a constraint would exclude precisely the mixtures studied below. Deterministic history-dependent policies and randomized history-dependent policies are kept separate. A pointwise lower bound for all these histories can be integrated against any initial distribution; an optimum computed at one fixed initial state cannot simply be integrated to claim a Markov optimum with common continuation choices.

Operating preservation relative to the installed rule, $J_t^\pi \geq J_t^{\pi^0}$, is checked separately. It is silently added to (3). In the primary experiment every selected deployment satisfies both properties, so a deployment with a matched lower bound also solves the problem with the additional preservation restriction.

2.1. Maintenance primitives and the unchanged primary cohort

The state $x \in [0, 1]$ measures condition. Actions 0, 1, 2 mean defer, maintain, and replace. The primitive parameters are

$$\beta = 19/20, \quad r(x, a) = x - c_a, \quad (c_0, c_1, c_2) = (0, 3/20, 9/20), \quad g(x) = x/2. \quad (5)$$

$$\begin{array}{c|cc}
a & f_{a1}(x) & f_{a2}(x) \\
\hline
0 & 3x/4 & 3x/4 + 1/20 \\
1 & 17x/20 + 1/10 & 17x/20 + 7/50 \\
2 & x/10 + 4/5 & x/10 + 17/20.
\end{array} \tag{6}$$

map preserves $[0, 1]$. The primary revision weight is $1 + x$, and ν is uniform. Because revision has a positive current operating cost, its justification comes from continuation value. Independent transitions also make future implementation exposure endogenous. The primary cohort is unchanged: horizons 4, 8, 12, tolerances $1/100, 1/20$, and seven frozen rules at each horizon. These are defer, an occupancy-stress rule, two spline rules, and three neural rules. The neural seeds are 31001, 31002, and 31003; the spline grids have 129 nodes. The exact serialized R34 incumbents, not refitted substitutes, are used. Source hashes and separate values at partition knots are part of the new proof objects. A floating-point network evaluation is not interchangeable with this exact compiled policy evaluation.

3. CONSTRUCTIVE WITNESSES AND FEASIBLE DEPLOYMENT

bounded functions L_t, U_t , suppose

$$L_t \leq V_t \leq U_t, \quad L_T = U_T = g. \tag{7}$$

inequalities must be constructed and checked, not supplied as a costless oracle. In the resource economy, expectation under positive affine maps, addition of affine rewards, and rounding preserve convex piecewise-affine structure. The exact representation stores rational coefficients in an affine formula on each open interval, and separate values at every knot.

LEMMA 1—Constructive upper encoding: *Let F be continuous, convex, rational piecewise linear on $[0, 1]$. Its chord interpolant on a rational mesh lies above F . Rounding the starting value and each chord slope upward to a common grid $2^{-q}\mathbb{Z}$ produces a convex function G with $G - H \leq 2^{1-q}$, where H is the chord interpolant. If F is M -Lipschitz and mesh width is at most h , then $0 \leq G - F \leq Mh + 2^{1-q}$. The actual maximum defect is determined exactly as the union of the two knot sets.*

PROOF: Convexity orders the chord slopes; upward rounding on a common lattice preserves convexity. Each slope error is below 2^{-q} , so integration over a total length of one adds less than 2^{-q} to the starting error. Convexity puts every chord above F , and the endpoint Lipschitz conditions bound the chord error by Mh . The difference of the two represented functions is bounded on each common interval, so its extrema occur at endpoint limits and stored points. *Q.E.D.*

Starting from $U_T = g$, construct $F_t = T_t U_{t+1}$ and an encoding satisfying $0 \leq U_t - F_t \leq d_t$. If $d_t = 0$ and $b_t = d_t + \beta b_{t+1}$, set $L_t = U_t - b_t$. Backward comparison proves (7). In the resource economy a slope bound propagates as $M_t \leq 1 + \beta(17/20)M_{t+1}$ before the explicit slope increment. Thus finite mesh and precision choices exist for every positive prescribed error. The implementation checks the actual defects rather than relying only on a priori error bounds.

$$\mathcal{A}_t(x) = \{a : U_t(x) - T_t^a U_{t+1}(x) \leq \eta_t\}. \quad (8)$$

nonempty whenever $d_t \leq \eta_t$. Every selector in this class satisfies $U_t = J_t^\pi \leq e_t$ by induction hence satisfies (3). Minimizing the implementation recursion over $\mathcal{A}_t$ solves the cost problem inside this sufficient class. It need not solve the unrestricted feasible problem. The implementation cost of a deployment is always evaluated using that deployment's transition law. A pointwise choice minimizing $k_t(x, a)$ inside $\mathcal{A}_t(x)$ discards the term $\beta P_t^a C_{t+1}$; dynamic choice retains it. Their comparison is meaningful only when both use the same admissible actions and each policy's own occupancy. Section 7.3 imposes both requirements. For upper-bound enrichment, a backward generator carries an actually constructed continuation (J_{t+1}, C_{t+1}) , admits only actions satisfying $V_t - T_t^a J_{t+1} \leq \varepsilon$, and selects among them using an operating-priority score. This generator uses exact V only in the reference study. Its weights 0, 1, 4, 16 are computational search parameters, not estimated economic prices. Any policy admissible set is a failed candidate. The selected full policy is reevaluated before use, and optimality claim is attached to the generator itself.

4. DETERMINISTIC OUTER BOUNDS AND OPTIMALITY CERTIFICATES

Define the witness-based necessary set and its cost recursion by

$$\mathcal{N}_t(L, U; x) = \{a : L_t(x) - T_t^a U_{t+1}(x) \leq \varepsilon\}, \quad (9)$$

$$B_T^{L,U} = 0, \quad B_t^{L,U}(x) = \min_{a \in \mathcal{N}_t(L, U; x)} \{k_t(x, a) + \beta P_t^a B_{t+1}^{L,U}(x)\}. \quad (10)$$

The set is nonempty because an operating-optimal action belongs to it. Unlike (8), this is an upper restriction: every feasible deterministic action must pass the test, but passing the test does not ensure that accumulated future loss remains feasible.

THEOREM 1—Safe deterministic outer bound: *Under (7), every feasible deterministic Markov policy satisfies $B_t^{L,U}(x) \leq C_t^\pi(x)$ at every restart. The same inequality holds for every deterministic history-dependent policy satisfying (4), with $x = x_t$ and its history-conditioned cost. If a separately feasible Markov policy $\hat{\pi}$ has*

$$\int B_0^{L,U} d\nu = \int C_0^{\hat{\pi}} d\nu, \quad (11)$$

then it satisfies $K_D(\nu, \varepsilon)$ and also the infimum over the stated deterministic history-dependent class. Equality of the complete functions $B_t^{L,U} = C_t^{\hat{\pi}}$ additionally establishes cost optimality from any restart. Integrated equality alone does not establish that stronger assertion.

PROOF: At a feasible deterministic decision history with current action a , operating comparison gives

$$L_t(x) - T_t^a U_{t+1}(x) \leq V_t(x) - T_t^a V_{t+1}(x) \leq V_t(x) - J_t^\pi(h_t) \leq \varepsilon.$$

The middle inequality uses optimal continuation as an upper bound on the policy's actual continuation after each successor history. Hence $a \in \mathcal{N}_t$. Assuming the cost inequality at the date, positivity of P_t^a gives

$$B_t^{L,U}(x) \leq k_t(x, a) + \beta P_t^a B_{t+1}^{L,U}(x) \leq C_t^\pi(h_t).$$

feasible initial objective. A feasible policy attaining it is optimal. For pointwise equality the argument applies separately at every restart. An integral can ignore other dates, zero-points, and states not cost-relevant under ν , so no converse pointwise implication is e. 22 *Q.E.D.*

theorem uses exact comparisons, not a floating tolerance for equality. More generally, a interval of width at most τ certifies τ -optimality for its named deterministic objective. a valid stopping criterion whenever attained; it is not a claim that every available ent must attain it.

POSITION 1—Refining witnesses: *If $L^n \uparrow V$ and $U^n \downarrow V$, then the sets $\mathcal{N}(L^n, U^n)$ contain $\mathcal{N}(V, V)$, and the associated bounds increase to $B^{V,V}$ pointwise. For a finite et, a uniform strict margin separating excluded actions from the threshold ensures of the necessary sets once the witness errors are smaller than that margin. Neither nt identifies $B^{V,V}$ with the constrained optimum unless a feasible upper policy closes val.*

F: The defining left-hand side increases with n . An action whose exact gap exceeds ntually excluded; an action whose exact gap is at most ε is always included. A finite et therefore stabilizes pointwise to the exact necessary set. By backward induction, ded nonnegative cost recursions increase; monotone convergence passes through each ion and a minimum over finitely many actions. The uniform-margin assertion follows $|V - T^a V) - (L^n - T^a U^n)| \leq \|V - L^n\|_\infty + \beta \|U^n - V\|_\infty$. The exact necessary test ws policies whose losses accumulate, which explains the final qualification. *Q.E.D.*

e experiment we compute both $B^{L,U}$ and $B^{V,V}$. The former demonstrates that safe ent bounds do not require an exact operating solution. The latter is a fully charged ject deterministic reference bound. Their difference is measured, not suppressed. When t-set minimizing selector itself is infeasible, it remains a valid lower-bound witness but deployable policy.

5. RANDOMIZATION AND DIRECT CONSTRAINED REFERENCES

nwise screening cannot be applied to randomized policies. A mixture can satisfy the g constraint while assigning positive probability to an action that fails the deterministic y test. We therefore retain the unrestricted support/restart bounds for the randomized d introduce a separate constrained reference calculation.

5.1. A continuation-frontier recursion

ite shocks $j = 1, \dots, m$, let $\mathcal{K}_t(x)$ be the achievable pairs of conditional operating value ementation cost from a decision history ending at (t, x) , with all future decision histories g (4). The dependence on the installed rule is suppressed. Put $\mathcal{K}_T(x) = \{(g(x), 0)\}$.

$$\mathcal{K}_t(x) = \text{co} \left(\bigcup_{a \in A} \mathcal{S}_t^a(x) \right) \cap \{(j, c) : j \geq V_t(x) - \varepsilon\},$$

$$\mathcal{S}_t^a(x) = \left\{ \left(r_t(x, a) + \beta \sum_j p_j j_j, k_t(x, a) + \beta \sum_j p_j c_j \right) : \right. \\ \left. (j_j, c_j) \in \mathcal{K}_{t+1}(f_{aj}(x)), \quad j = 1, \dots, m \right\}.$$

the efficient lower frontier is needed: higher cost at the same or lower operating value dominated. At a fixed restart, the minimum second coordinate of (12) is the exact history-conditioned randomized implementation optimum.

THEOREM 2—Finite constrained frontier: *For a finite horizon, finite actions, and finite shocks, recursion (12) is exact for the history-conditioned class. Rational primitives and rational state transitions give finite rational polygonal frontiers. Primal convex combinations and supporting lines certify every retained frontier edge and its minimum cost. The recursion need not compute randomized Markov optimum when different histories reaching the same physical state may use different continuations.*

PROOF: At the terminal date the assertion is immediate. Conditional on a current action and next shock, the policy must choose a continuation pair in the corresponding successor set. The expectation of these pairs is exactly the action-specific set in braces. Mixing current sets produces its convex hull. The half-space enforces current feasibility; successor sets already enforce every subsequent decision-history restriction. Conversely, choose a convex representation, implement its current lottery, and use the selected feasible successor continuation for each realized action and shock. This constructs every pair in the recursion. Finite Minkowski sums, finite unions, convex hulls, and half-space intersections preserve rational polytopes. Supporting lines bound all candidate vertices and primal combinations attain the retained edges; adjacent supporting slopes certify entire edges. The construction remembers which continuation was assigned to each history and imposes no common Markov continuation across distinct histories. *Q.E.D.*

The full tree can grow exponentially with horizon. Memoization of identical (t, x) frontier reduces repeated construction, but it does not impose equality of the selected continuation sets at separate visits. For continuous shocks, additional validated set approximation would be required. We do not infer a high-dimensional complexity rate from a finite tree.

The reference study solves 70 fixed-restart horizon-four problems: the 14 primary configurations at $x = 0, 1/4, 1/2, 3/4, 1$. These objectives are Dirac initial distributions, not the primary norm distribution. In 24 cases the randomized history-conditioned value is strictly below the deterministic lower bound at the same restart. Each full frontier has a separately implemented primal/dual verification. The experiment is a direct constrained comparator for its exact policy class and initial state, not a numerical quadrature of K_R in the primary study.

5.2. A solved randomized Markov continuum example

The two-period version yields an exact comparison under the *same uniform initial objective* for both policy classes. Keep all maintenance primitives and install defer at both dates. At the decision date defer maximizes operating value and has zero implementation cost. Thus it

0, 1/20, exact partition evaluation gives $\Delta(x) > \varepsilon$ throughout $[0, 1]$. A feasible deterministic policy must then change its first action at every state, so $K_D = \int_0^1 x dx = 3/2$. For a randomized policy, all changed actions have the same implementation operating²⁴ maximizing changed action produces the largest gain per intervention. The probability of changing is consequently $1 - \varepsilon/\Delta(x)$, and

$$K_R = \int_0^1 (1+x) \left(1 - \frac{\varepsilon}{\Delta(x)}\right) dx < \frac{3}{2}. \quad (13)$$

A feasible Markov policy at every initial state; its last-date continuation is defer and is at every restart. Any mixture with a smaller change probability would lose more than ε g value at that state. Equation (13) therefore solves the randomized Markov problem, only a local necessary relaxation.

An affine piece $\Delta(x) = ax + b$, the integral of $(1+x)/(ax+b)$ equals $(r-l)/a + \log((ar+b)/(al+b))$ when $a \neq 0$, with the evident constant-denominator formula. Exact rational logarithm enclosures yield costs approximately 1.3720075152810114 500375764050574. The enclosure widths are below 10^{-20} . At $x=0$, $\Delta = 2323/4000$ optimal replace probability at tolerance $1/100$ is 2283/2323. This counterexample rules out deterministic sufficiency for the model. It supplements rather than replaces the unchanged horizons.

6. RANDOMIZED LOWER BOUNDS AND ADAPTIVE LOCAL CERTIFICATES

global nonnegative support price λ , let $W_T^\lambda = \lambda g$ and

$$W_t^\lambda = \max_a \{ \lambda r_t(\cdot, a) - k_t(\cdot, a) + \beta P_t^a W_{t+1}^\lambda \}. \quad (14)$$

A feasible randomized or history-conditioned policy satisfies $W_t^\lambda \geq \lambda J_t^\pi - C_t^\pi$, hence

$$F_t(x) = \max \{ 0, \sup_{\lambda \in \Lambda} [\lambda(L_t(x) - \varepsilon) - W_t^\lambda(x)] \} \leq C_t^\pi(h_t). \quad (15)$$

Λ gives a valid floor without claiming that its supremum exhausts global duality.

R_{t+1} be a cost lower bound valid after every successor decision history. Set $q_a = L_t(x)$, $z_a = k_t(x, a) + \beta P_t^a R_{t+1}(x)$, and $b = L_t(x) - \varepsilon$. Feasibility implies $\sum_a p_a q_a \geq b$. Continuous local relaxation is

$$\Phi_t(R_{t+1})(x) = \min_{p \geq 0, \sum p_a = 1, \sum p_a q_a \geq b} \sum_a p_a z_a = \sup_{\mu \geq 0} \min_a \{ z_a + \mu(b - q_a) \}. \quad (16)$$

Support price λ in (14) prices a complete unrestricted continuation. The local multiplier μ only the necessary one-step inequality in (16). They are not interchangeable.

PROPOSITION 2—Local restart propagation: *If F_t and R_{t+1} are valid cost lower bounds for every-history-conditioned feasible class, then $R_t = \max\{F_t, \Phi_t(R_{t+1})\}$ is also valid. For three the pointwise primal optimum is attained either by a feasible pure action or by a mixture of actions straddling b . The dual optimum is attained at zero or at a nonnegative intersection of action lines, with ties and equal slopes handled explicitly.*

al continuation costs dominate R_{t+1} . Their cost therefore dominates the minimum in (16). The maximum with another valid lower bound is safe. The simplex intersected with half-space has vertices of the stated type. The dual is a concave piecewise-linear function $(0, \infty)$; feasibility and bounded primal cost imply a line contained in the simplex and an intersection or on a constant segment, including zero when applicable. Direct enumeration of the candidates supplies the primal and dual equality. *Q.E.D.*

on an interval where q_a, z_a, b are affine, a two-action primal cost is

$$P_{ij}(x) = \frac{(q_j - b)z_i + (b - q_i)z_j}{q_j - q_i}, \quad q_i \leq b \leq q_j. \quad (17)$$

numerator is quadratic and its denominator affine. A fixed dual price gives $D_\mu(x) = \mu\{z_a + \mu(b - q_a)\}$, a piecewise-affine lower bound. We first split at all input knots, eligibility crossings, and denominator sign changes. A cell is accepted only if the primal is feasible throughout it and $P_{ij} - D_\mu \leq \rho/2$ throughout it. Multiplication by the nonnegative denominator reduces each action-line comparison to a quadratic sign test. A quadratic's maximum on a closed interval occurs at an endpoint or its interior stationary point. Zero denominators at cell endpoints are handled by endpoint limits; stored isolated knots are solved separately by the exact point LP.

THEOREM 3—Certified local approximation: *Suppose the preceding cell tests cover the action. Rounding every coefficient and isolated point value of the dual lower function downward by dyadic bits decreases it by at most 2^{1-q} on $[0, 1]$. Let $e = \rho/2 + 2^{1-q}$ and assume $2^{1-q} \leq R_t$. Backward propagation with this lower encoding, and the same exact floor F_t , produces R_t with*

$$0 \leq R_t - \hat{R}_t \leq e \sum_{s=0}^{T-t-1} \beta^s. \quad (18)$$

finite affine partitions with a strictly feasible action margin on each compact cell, refinement terminates for every positive ρ . At degeneracies, finite termination is asserted only when the primal one-sided extension and cell tests admit a finite cover. A refinement cap causes an explicit failure, never a passing certificate.

PROOF: For $x \in [0, 1]$, decreasing slope and intercept by less than 2^{-q} decreases an affine value by less than 2^{1-q} . Stored point values have the same direction. The cell test bounds the distance to the exact local primal optimum before rounding. Changing a continuation cost by at most d changes every action cost by at most βd , so minimization over the unchanged feasible simplex is monotone and β -nonexpansive. The maximum with the same floor is nonexpansive. Backward induction gives (18). Under a strict feasibility margin, feasible primal vertices and primal dual choices have bounded local moduli on a finite subdivision; the midpoint primal/dual error then tends uniformly to zero as cell widths shrink. Compactness supplies a finite cover. At boundary degeneracy such a modulus need not follow from strict interior feasibility alone, which is why the stated one-sided condition and explicit stopping test are required. *Q.E.D.*

The initial unquantized reconstruction encountered rational serialization growth. That unsuccessful run is retained. The implemented experiment uses $q = 40$ and $\rho = 1/100, 1/200, 1/400$, and directed 64-bit dyadic integration enclosures. It solves the two horizon-four neural31001 distance cases under constructed witnesses and exact V , for 12 complete certificates. The floor

Horizon	Cases	Exact initial	Positive exact	All-restart exact	Max. gap
4	26	14	10	14	0.000000
8		6	2	5	0.103473
12		4	0	4	0.248472

Exact initial means equality of rational uniform-initial costs. All-restart exact additionally means equality of exact cost-to-go function. Both operating tolerances are included. The gap is a deterministic cost interval, not a duality gap.

at zero throughout this diagnostic. The experiment measures approximation to this local relaxation, not an additive decomposition of a different nonzero-support restart and not a proof of global strong duality.

7. NUMERICAL EVIDENCE AND COMPUTATIONAL ACCOUNTING

new results are generated from executable sources and exact or outward-directed proof. The earlier R35/R36 protocol assertions are not used as observations. The primary model, installed configurations, and its operating tolerances are unchanged. The computation specifies commands, source hashes, execution records, arithmetic libraries, and the unsuccessful candidate ledger.

7.1. Primary deterministic optima and residual intervals

Exact necessary-set lower recursion and independently feasible deployments agree in 24 initial objectives, including 12 strictly positive costs. Ten positive-cost equalities occur on four and two at horizon eight. The horizon-eight neural31003 case with $\varepsilon = 1/20$ has exact equality without equality of all cost-to-go functions. The other 23 exact cases have function equality. This distinction is tested, not inferred from rounded table entries. Horizon-four necessary selectors are feasible, and their full cost functions match those selected feasible deployment. This supplies a certificate-level explanation for all 14 : in these configurations the outer relaxation itself has a feasible minimizing selector. a theorem that every horizon-four maintenance specification has this property. At horizons the necessary selector violates the operating target in 19 cases, by as much as nominally 0.07738. Eighteen of those retain a positive initial cost interval; the remaining the integrated-only equality just described. Accumulated operating loss, not failure of arithmetic, is the diagnosed obstruction to treating those outer selectors as deployments. primary deployment passes the all-state operating constraint and preserves the installed g value. Both candidate failures and all 18 residual deterministic intervals are printed in element or recorded exactly in the result ledger. The lower and upper fractions, policy er-function hashes, frozen-input hash, policy class, objective, and exactness scope are l for each of the 42 cases. Decimal displays do not determine equality.

7.2. Controlled gap diagnostics

support floors are compared under identical installed rules: constructed L with eight prices, exact V with the same eight prices, and exact V with fifteen prices. The eight re 0, 1, 4, 16, 64, 256, 1024, 4096; refinement adds $1/2, 2, 8, 32, 128, 512, 2048$. This re-15 fresh unrestricted support solves across 21 horizon-incumbent pairs; both operating

Coordinate	Strict improvements	Largest gain
Support witness	30	0.00008591
Global price refinement	28	0.31597630
Deterministic witness	30	0.00001170
Feasible upper enrichment	30	0.93987965

Each row compares identical initial distributions and installed rules within its own policy-class scope. Gains across rows are not additive. Global prices expand from eight to fifteen; the local LP study has its own zero floor.

TABLE III

CONTINUOUS LOCAL RESTART RELAXATION: EXACT-VALUE DIAGNOSTIC

ε	ρ	Eight-price lower	Adaptive lower	Adaptive upper
0.01	1/100	0.076186	0.088078	0.106627
0.01	1/200	0.076186	0.088107	0.097382
0.01	1/400	0.076186	0.088121	0.092759
0.05	1/100	0.020289	0.025620	0.044169
0.05	1/200	0.020289	0.025649	0.034924
0.05	1/400	0.020289	0.025665	0.030302

Horizon four, frozen neural31001, zero support floor, 40-bit directed coefficients. The endpoints enclose the continuous local relaxation, not the global constrained optimum.

These refinements reuse the same support functions. Pointwise monotonicity of both refinements is checked.

The witness substitution improves 30 support floors, with maximum integrated gain below 10^{-5} . Global price refinement improves 28 floors, with maximum gain approximately 0.316. In the separate deterministic ladder, exact V raises 30 safe outer bounds, with maximum gain below 1.2×10^{-5} . Upper-portfolio enrichment lowers 30 inherited deployment costs, by as much as approximately 0.9399. These measurements show why exact-value availability is not the main numerical source of the residual deterministic gaps on this cohort.

These gains are not summed across policy classes. Global support floors are valid for randomized history-conditioned policies; the necessary-action floor is deterministic. The local LP diagnostic has zero floor and uses only the two declared horizon-four neural31001 cases. For 18 unresolved primary cases, the exact deterministic interval, necessary-selector violation, best gain, support refinement, and upper-search improvement are reported as separate coordinates. We do not claim that the structural gap of the continuous global randomized problem has been numerically isolated.

Tightening the local tolerance narrows a certified interval around the continuous local relaxation. The certified comparison with the eight-price local recursion is computed on identical test inputs and a zero support floor. The theorem explains the narrowing without assigning it to global strong duality.

7.3. Economic sensitivities and endogenous exposure

The sensitivity study uses horizon eight, three natural installed rules, two separately declared price ranges 0.05 and 0.20, and eleven one-at-a-time primitive specifications. The baseline control rule replaces below 0.3 and maintains below 0.6. The preventive rule replaces below

Installed rule	Cases	Strict savings	Zero savings	Largest saving
condition	22	2	20	0.015754
preventive	22	16	6	0.042212
calendar	22	4	18	0.087011

h policies use identical sufficient action restrictions and their own transition occupancies. Eleven one-at-a-time modifications and two operating tolerances generate 22 cases per installed rule. These counts are not population means.

l maintains below 0.75. The calendar rule intervenes every fourth date, replacing below maintaining otherwise, and defers at other dates. These are normalized scenarios, not transitions sampled from firms.

primitive perturbations change discounting to 0.90 or 0.99, the revision weight to 1 or the initial density to $2x$ or $2(1-x)$, maintenance cost to 0.10 or 0.20, or defer persistence or 0.85. Only the named primitive changes in each scenario. Both dynamic and pointwise use the same sufficient action sets; their costs are evaluated under their own transitions same scenario-specific initial density. All 66 configurations pass their operating checks. The 22 strictly positive implementation savings and 44 exact zero effects; the maximum savings approximately 0.08701. The complete scenario table retains the zero effects. The demonstrates sensitivity of intervention exposure across the declared economic rules, valence estimate or a dollar welfare calibration. Computational penalties used elsewhere substituted for these economic revision weights.

7.4. A nonlinear two-state economy

tion is now $(x, y) \in [0, 1]^2$, and actions defer, repair the first component, repair the component, or overhaul both components. Rewards are $(x + y)/2 - c_a$, with costs (0.18, 0.32), terminal payoff $(x + y)/4$, and revision weight $1 + (x + y)/2$. Discounting check probabilities remain 0.95 and (0.6, 0.4). Each coordinate transition has the form $y + c + n\xi)/100$, where $\xi \in \{0, 1\}$ and z is that coordinate. The action-specific tuples

	x -coordinate	y -coordinate
0	(65, 5, 0, 3)	(70, 3, 0, 2)
1	(15, 3, 70, 2)	(70, 3, 0, 2)
2	(65, 5, 0, 3)	(15, 4, 70, 2)
3	(10, 2, 80, 2)	(10, 2, 80, 2)

maps are monotone on the square and preserve it, but bilinearity destroys the one-dimensional affine closure used by the primary solver.

firm meshes have $N = 16, 32, 64, 128, 256$ cells per coordinate, and horizons are 32. Box endpoint arithmetic and backward floors/ceilings are performed with signed at scale 2^{34} . The code encloses each full transition image, enumerates every intersected box, and checks an intermediate absolute-value bound below 2^{62} . The grid policy is on each assigned half-open box; closed transition enclosures cover ties conservatively. including separately evaluated fixed policies, appear in the supplement.

operating tolerance is 1/2 in this separate economy. It does not replace 1/100 or 1/20 primary study. Two restart-based generators use penalties zero and four. A classical

Horizon	Restart generators	Classical portfolios	Compiled-rule pilot
4	9/10	5/5	0/4
8	7/10	4/5	0/4
16	6/10	3/5	0/4
32	4/10	3/5	0/4

. Each entry is certified/attempted. The separate operating tolerance is $1/2$. Restart and classical methods use meshes 16 through 256; the compiled-rule pilot uses meshes through 128. The two restart penalties are separate outputs, two independent instances of one comparator.

TABLE VI
CHARGED PRIMARY CONSTRUCTION WORK AND REPRESENTATION

Horizon	Operating DP (s)	Median revision (s)	Max. lower pieces	Max. bits
4	0.0046	0.0973	11	143
8	0.0141	0.3478	90	186
12	0.1198	2.3529	430	242

. The operating DP is executed once per horizon. Revision time includes both exact-value and safe-witness outer bounds and the complete upper portfolio. Independent verification, support diagnostics, and supplementary studies are optional work, not omitted overhead. Timings are generated from this publication execution.

near-interpolation grid solver constructs a scalarization portfolio at prices 4, 16, 64, 256; its stopping-point output is only a proposal and receives the same directed whole-box evaluation. Witness, candidate, and verification work is charged. There are 26 certified restart-generator configurations among 40 and 15 certified classical portfolios among 20. A fresh R38 audit of the point-compiled installed-rule box policy fails in all 16 pilot mesh–horizon configurations through mesh 128. That compiled-rule pilot is not a reconstruction of the unsupported R35/R36–penalty pilot claim. The two algorithms’ candidate counts differ, so their success counts are a paired win rate. The full ledger reports each matched mesh, horizon, tolerance, runtime, enclosure, witness width, fallback count, and failure. These results demonstrate executed linear enclosure construction, not uniform dominance over the classical comparator.

7.5. Time, representation, and verification

Exact operating solves are shared only across configurations with the same horizon. Frozen compiled rules are amortized historical inputs; no neural training is described as newly executed. Exact operating values, outer recursions, feasible candidate searches, support refinements, tier references, adaptive LP construction, nonlinear grids, and their verification are charged separately. This distinction avoids comparing warm certificate work with cold neural training.

SymPy Rational verifier imports none of the Fraction/PW construction, envelope, composition, or optimizer routines. It rebuilds partitions using actual primitive transition pullbacks, checks necessary-set threshold crossings, open-cell endpoint limits, isolated values, Bellman utilities and inequalities, all-restart feasibility, terminal conditions, and exact initial integrals. 42 primary objects pass. Twenty declared corruption categories are rejected, including positions, discounting, shock probabilities, action labels, parser failures, objective scope, point sets, and false exactness claims.

A second SymPy procedure independently checks the 70 fixed-restart frontier objects using explicit primitive candidate pairs, feasible convex combinations, and supporting inequalities.

tor's arithmetic kernel. Its coverage is not extended to the nonlinear arrays or the two-argument calculation. The local cell checker and the nonlinear reevaluator are useful but their respective construction arithmetic; we do not call those implementations independent. A common mode risks include a common interpretation of the model and errors in duplicated primitive specifications. Canonical hashes identify the checked objects, not of an unchecked mathematical assumption.

8. STOPPED ECONOMIC CONTROL AND A KILLED-OPERATOR BRIDGE

original consumption–portfolio program is retained with its original target. On $(u, x) \in (0, 1) \times (0.5, 2)$ and horizon one,

$$dZ_s^u = \theta_s ds + 0.05 dZ_s^u, \quad (19)$$

$$d[(0.02 + 0.06p_s)x_s - c_s]ds + 0.2p_s x_s dZ_s^x, \quad d\langle Z^u, Z^x \rangle_s = -0.25 ds. \quad (20)$$

range over $[0.05, 0.8] \times [-0.2, 0.2] \times [-0.5, 0.8]$. Running utility is $c^{1-u}/(1-u) - \text{terminal payoff is } G = -0.02(u-2)^2 + 0.1 \log x$, and exit settlement is $G - 8(1-s)$, discount rate 0.04. The full current-state target at $k=2$ remains $\sup_{t,u,x} (V_2 - J_2^\pi) \leq 0.01$. Connection is operator-theoretic rather than a claim that maintenance certificates solve the problem. Let τ be exit, Δ a time step, and a a control held over the step. Define the killed operator

$$\begin{aligned} \mathbb{T}_\Delta^a w(z) = \mathbb{E}_z^a \left[\int_0^{\tau \wedge \Delta} e^{-rs} \ell(Z_s, a) ds + e^{-r\tau} g(t + \tau, Z_\tau) \mathbf{1}\{\tau \leq \Delta\} \right. \\ \left. + e^{-r\Delta} w(Z_\Delta) \mathbf{1}\{\tau > \Delta\} \right]. \end{aligned} \quad (21)$$

continuation measure is positive and has mass at most $e^{-r\Delta}$. Bellman comparison, witness criterion, and the necessary-action proof therefore extend verbatim to this subprobability measure with the continuation mass replacing β and exit settlements included in the one-step problem. A verified operator error ξ_t enters the backward witness budget additively, and thus contributes at most the corresponding discounted sum of ξ_t to the operating certificate.

This statement is useful only with valid killed-transition and action-coverage enclosures. A fine action grid alone cannot supply a lower bound for the original continuous-control revision problem: omitted actions may improve it. Uniform action interpolation, time discretization, boundary settlement, and diffusion approximation errors must all be covered. Markov-chain approximation provides one route to such an additional analysis (Kushner and Dupuis, 2001); it is applied by the maintenance experiment.

The retained scalar boundary result remains intact. At restart $(0, 61/50, 5/4)$ with $k=2$, $p=0$, and constant $\theta \in [-1/5, 1/5]$, the archived derivative enclosures are positive on the edges of the interval and identify $\theta = 1/5$ as the unique maximizer in that scalar class. The cost and payoff enclosures are reproduced in the supplementary preservation material. The full-domain bound 7.181834580823298 is not converted into a 0.01 certificate. The killed-operator result explains precisely which parts of the certificate method transfer and which additional validated computations remain necessary for the unchanged diffusion objective.

operating feasibility at every restart and implementation optimality under an initial distribution can be certified separately and then combined. Safe necessary-action restrictions convert operating witnesses into deterministic cost lower bounds without requiring exact operating costs. On the unchanged primary cohort, matching feasible deployments establish 24 exact deterministic initial objectives, including 12 positive implementation costs. The separate randomized references show why a policy-class qualifier is mathematically indispensable, rather than a presentational qualification.

The adaptive local LP supplies a controlled approximation to a continuous restart relaxation. Error bound, the global support-price study, and the remaining deterministic intervals identify current numerical tasks rather than hiding them in a single gap. Nonlinear whole-box execution and the designed economic sensitivities extend the evidence beyond the exact affine reference level. The publication object includes the proofs, source, unsuccessful computations, and actual verification records needed to review these claims. The original stopped-control objective and earlier technical material are preserved with an explicit map to their present location.

REFERENCES

- Altman, E. (1999): *Constrained Markov Decision Processes*. Chapman and Hall/CRC. [2]
- Boyd, S. P., and J. E. Smith (2014): “Information Relaxations, Duality, and Convex Stochastic Dynamic Programs,” *Operations Research*, 62(6), 1394–1415. doi:10.1287/opre.2014.1322. [2]
- Boyd, S. P., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58(4, part 1), 785–801. doi:10.1287/opre.1090.0796. [2]
- Chen, J., and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Biometrika*, 85(5), 1575–1612. [2]
- Rockafellar, T. A. (1998): *Numerical Methods in Economics*. Cambridge, MA: MIT Press. [2]
- Kushner, H. J., and P. G. Dupuis (2001): *Numerical Methods for Stochastic Control Problems in Continuous Time*, second edition. New York: Springer. [14]
- Leventhal, J., P. Trichelair, and R. Tachet des Combes (2019): “Safe Policy Improvement with Baseline Bootstrapping,” *Proceedings of Machine Learning Research*, 97, 3652–3661. [2]
- Os, R., and C. Szepesvári (2008): “Finite-Time Bounds for Fitted Value Iteration,” *Journal of Machine Learning Research*, 9(27), 815–857. [2]
- Rockafellar, T. A. (2010): “Verification Methods: Rigorous Results Using Floating-Point Arithmetic,” *Acta Numerica*, 19, 37–449. [2]
- Rockafellar, T. A. (2013): “Performance Bounds for λ Policy Iteration and Application to the Game of Tetris,” *Journal of Machine Learning Research*, 14(36), 1181–1227. [2]

12. VERBATIM PRESERVED R38 TECHNICAL SUPPLEMENT

This supplement gives the representation, policy-class, constrained-frontier, cell-verification, and directed-box details supporting the article. It reports every primary configuration, every economic sensitivity, every fixed-restart reference, all nonlinear failures, and all adaptive local LP runs. The original stopped-control scalar proof is retained in full. Exact fractions and complete proof objects accompany the displayed numerical summaries.

1. POLICY CLASSES, ATTAINABILITY, AND RESTART SEMANTICS

The deterministic Markov feasible class in the article uses a single measurable action at each date and state. Its randomized counterpart uses a probability kernel at that date and state. The restart condition is imposed before the current action draw. Neither class is replaced by an all-state average performance constraint. Implementation cost is averaged under the specified initial distribution only after the all-restart operating inequalities have been established.

A history-conditioned policy may remember earlier actions, shocks, and previously drawn randomization. Its continuation is evaluated at a decision history before the next action draw. To make off-path assertions well-defined, a policy includes continuation kernels at every admissible history, not only histories with positive probability under the initially deployed policy. The operating benchmark remains the unconstrained Markov value $V_t(x_t)$ because the underlying model is Markov. The required inequality is conditional at every such history.

The deterministic necessary-action proof uses the action actually selected at that history. Its operating continuation is V_{t+1} , regardless of the particular history. A randomized policy instead obeys an average inequality over current actions; individual actions in its support need not obey the deterministic test. The two-period example is an explicit instance. Conditioning on the current action draw would define a different feasible class and would remove the mixture mechanism; that conditioning is not used here.

The exact frontier recursion keeps separate continuation choices for different histories that end in the same physical state. Identical state/date frontier sets can be cached because the primitive model is Markov. Selecting two different points on the cached frontier at different histories is not history dependence. Thus memoization is not a proof that a randomized Markov policy attains the frontier optimum.

At a fixed initial state with finite shocks and finite horizon, the reachable decision tree is finite. A feasible solution on that tree can be extended to unvisited decision histories by an operating-optimal policy. The off-tree extension has finite implementation cost and zero operating regret. It does not affect the Dirac initial objective. This justifies the fixed-restart reference under the extended all-history definition without claiming that independently chosen references at different initial states admit a common Markov continuation.

2. EXACT FINITE REPRESENTATION AND INDEPENDENT CHECKING

The represented function is the finite tuple (z_i, m_i, b_i, v_i) , equal to $m_i x + b_i$ on each open interval (z_i, z_{i+1}) and to v_i at z_i . The terminal and operating witnesses are continuous in the

may affect later values through a transition preimage. Such points are not discarded because the initial distribution is continuous.

fine transition preimages of continuation knots are inserted. Necessary-set thresholds are also inserted before checking an action's eligibility. An inequality between affine functions on an open cell is equivalent to inequalities at both endpoint limits. Actual endpoint checks are checked separately. A midpoint identifies the correct affine formula on an already partitioned cell; it is not the test of the inequality itself.

independent primary verifier uses SymPy Rational and its own tuple parser, line selection, union, transition pullback, and primitive Bellman evaluator. It does not import the conda Fraction/PW kernel. For each object it checks the primitive model against duplicated the frozen installed-input hash, the terminal equations, $V = TV$, the upper-witness and lower-witness recursion, necessary-action masks, lower-cost minimum and attainment-deployed operating and cost equations, all-restart regret, and exact initial integrals. It updates both integrated and functionwise equality flags.

twenty mutation categories include primitive discount, transition, shock probability, cost, policy class, objective scope, terminal conditions, missing dates, knot order, malformation tokens, illegal actions, fractional actions, isolated values, affine coefficients, false equality, witness defects, and provenance/hash failures. The complete category and diagnostic outcomes are in the independent verification ledger. The independence concerns the separate arithmetic/partition implementation. Both programs intentionally omit the same economic primitives; this is not formal proof-assistant verification or an assumption that all common interpretation errors are impossible.

exact publication comparison also hashes canonical JSON for 187 scientific objects: 42 objects, 70 fixed-restart frontiers, 12 local LP objects, 21 global support bundles, and function objects. This cross-environment equality check is distinct from independent verification. Only the primary and frontier groups are covered by the independent SymPy hashes. Timing fields are not part of these 187 canonical scientific objects.

3. WITNESS CONSTRUCTION, SUFFICIENT ACTIONS, AND OUTER RESTRICTIONS

completeness, let $U_T = g$ and suppose $0 \leq U_t - T_t U_{t+1} \leq d_t$. Induction gives $V_t \leq U_t$. If $b_t = d_t + \beta b_{t+1}$, then

$$U_t - V_t \leq d_t + T_t U_{t+1} - T_t V_{t+1} \leq d_t + \beta b_{t+1} = b_t.$$

$b_t = U_t - V_t$ is a lower witness. Actual realized defects can produce a tighter witness uniform construction allowance. The R38 safe outer calculation uses the realized defect recorded in its proof object.

sufficient action class, suppose $e_t = \eta_t + \beta e_{t+1}$ and an action satisfies $U_t - T_t^a U_{t+1} \leq \eta_t$. Actual continuation satisfies $J_{t+1} \geq U_{t+1} - e_{t+1}$, then

$$U_t - T_t^a J_{t+1} \leq \eta_t + \beta e_{t+1} = e_t.$$

ing exclusively from nonempty sufficient action sets therefore gives an operating-feasible Cost minimization over those sets is ordinary dynamic programming, but it solves a r-class problem. In contrast, every feasible deterministic action is included in $\{a : U_{t+1} \leq \varepsilon\}$. The cost minimum over this latter class is a lower bound on the original deterministic problem.

er bound; it shows that the outer class is strictly larger than the operating-feasible class. Each
 uted feasible upper policy is evaluated from scratch. The main result’s equality test uses
 t rational integrals, and the stronger all-restart flag additionally checks every cost function
 erence on open cells and stored **COMMON-CONTINUATION CERTIFICATES: SUPPLEMENT**
 he upper-policy attempts per configuration are not presumed all feasible: the inherited
 oyment, necessary-set selector, operating-optimal selector, and four operating-priority
 ward selectors give seven named attempts. An empty backward admissible set is explicitly
 rded. The count is seven, not a number of independently optimized global policies. A
 lidate’s failed operating certificate is not silently repaired by increasing the tolerance.

4. EXACT POINT LP AND WHOLE-CELL PRIMAL–DUAL TESTS

t a fixed state the local LP is $\min p \cdot z$ subject to $p \geq 0$, $\sum p = 1$, and $p \cdot q \geq b$. Every vertex
 ther a feasible unit vector, or a mixture of two actions i, j with $q_i < b < q_j$ and weights

$$p_i = \frac{q_j - b}{q_j - q_i}, \quad p_j = \frac{b - q_i}{q_j - q_i}.$$

en $q_i = b$, the feasible unit vector handles the endpoint. When $q_i = q_j$, the pair does not
 ire division and its cheaper feasible unit vector or another crossing pair suffices. Ties are
 lved by a deterministic order, but every tied optimal value remains valid.
 he dual candidates are zero and nonnegative values $(z_i - z_j)/(q_i - q_j)$ for unequal q_i, q_j .
 implementation checks exact primal–dual equality at every sampled basis-selection point
 every stored isolated knot. These exact point checks select a certificate; they do not verify a
 by sampling.

n an open input cell, all q, z, b are affine. Additional splits ensure that action feasibility and
 ominator signs do not change in the interior. The proposed two-action primal has nonnegative
 ghts summing to one throughout the cell; these conditions are checked at endpoint limits.
 its numerator be N and denominator $D > 0$. For a fixed $\mu \geq 0$, write $d_a = z_a + \mu(b - q_a)$.
 e $D_\mu = \min_a d_a$, the inequality $P - D_\mu \leq \delta$ is equivalent to

$$N - D(d_a + \delta) \leq 0 \quad \text{for every action } a.$$

n left-hand side is quadratic. Its maximum is calculated using both endpoints and, for a
 ave quadratic, the vertex when interior. A zero denominator at an endpoint is treated by the
 nal one-sided extension and a separate exact point LP at the endpoint. A denominator that
 egative in the interior rejects the cell.

he piecewise-affine dual minimum can change active action within an accepted cell. All such
 ections are inserted, and the checker verifies that each stored affine line is the corresponding
 minimum throughout its subcell, not only at its midpoint. The union of accepted cells must
 er $[0, 1]$ without gaps or overlaps. The exact knot LP value is stored independently of the
 n-cell lower formula.

fter construction, each slope, intercept, and point value is rounded downward to a denom-
 or 2^{40} . On $[0, 1]$, an affine slope/intercept rounding changes the value by at most 2^{-39} in
 correct direction. The exact difference is checked again before propagation. For arbitrary
 uested tolerance the theorem requires enough bits to keep this rounding term below half the
 e budget; forty bits are the implemented choice for the three displayed tolerances, not a
 ision-free limit.

ite rational bounds avoids forcing every breakpoint denominator into one enormous common multiple. The width of this numerical integral enclosure is recorded separately and to the upper endpoint of the local-relaxation interval. No unrounded decimal integral is a proof. 36

initial unquantized reconstruction reached Python’s rational-to-string size limit. Its failure reserved, and the source labeled as a reconstruction is not claimed to be a bitwise copy of that initial process. The quantized algorithm and the final whole-cell verifier are executed sources. An engineering recursion-depth cap of 60 causes explicit failure. The final finite-cover theorem is not a promise that this cap suffices for arbitrary models.

5. FRONTIER PRIMAL AND DUAL VERIFICATION

constructor enumerates the action-specific Minkowski sums of efficient successor vertices, is the convex lower hull, and intersects it with the required operating half-space. Pruning is valid only when another achievable pair has no smaller operating value and no larger when the point lies above a supporting lower edge. Because feasibility is a lower on operating value, a higher-value cheaper pair dominates for all preceding lower-budget points.

every retained frontier vertex, the proof stores either one candidate or a convex combination of two candidates that attains it. Its cost-support line $c = \mu j + \eta$ is checked against primitive candidate pairs. Supporting slopes are nonnegative. Adjacent vertices determine the slope; checking that line against every candidate proves the entire edge, including the section with the operating-budget line. At the first feasible frontier point, either the budget binds or a zero supporting slope certifies an unconstrained cost minimum. The operating frontier coordinate is also checked against the independent operating value. The SymPy verifier reconstructs all candidate action/child combinations from the primitive and the already verified child frontiers. It compares canonical candidate counts/hashes and supporting inequality. It does not import the constructor’s hull or optimization functions. All nodes have the singleton $(g, 0)$. These tests certify the serialized recursive frontier, giving feasibility propagation, rather than just its reported initial number.

two-period uniform-initial randomized Markov calculation uses a different analytic site. On each rational interval the integrand is rational with an affine denominator. For a rational logarithm argument r , range reduction writes $r = 2^m s$ with $1 \leq s < 2$, using parity when $r < 1$. With $y = (s - 1)/(s + 1)$,

$$\log s = 2 \sum_{i=0}^{n-1} \frac{y^{2i+1}}{2i+1} + R_n, \quad 0 \leq R_n \leq \frac{2y^{2n+1}}{(2n+1)(1-y^2)}.$$

the expansion encloses $\log 2$. Forty terms give the recorded rational integral enclosures. The antiderivative coefficients determine which endpoint is used. This analytic calculation is not included in the count of 70 independently verified frontier objects.

6. NONLINEAR WHOLE-BOX ARITHMETIC AND CERTIFICATE SCOPE

box be $[i/N, (i+1)/N] \times [j/N, (j+1)/N]$ for range bounding. The represented policy is half-open assignment, with the upper outside boundary included in the final box. All coefficients in the nonlinear experiment are nonnegative. Therefore the lower/lower and upper/upper corner enclose the image of the entire box, including the bilinear term.

$$shN + bij + (c + n\xi)N^2,$$

COMMON-CONTINUATION CERTIFICATES: SUPPLEMENT.
denominator $100N^2$; the upper numerator replaces h, i, j by $h + 1, i + 1, j + 1$. Multi-
plication by N and integer division locate the possible successor boxes. The implementation
ensures that the index range spans at most one step in each coordinate, so evaluating all four
combinations of lower/upper indices covers every possible successor box. An exact upper image
grid boundary includes the neighboring assigned box, conservatively. It is not rounded into
single midpoint successor.

Rewards, terminal values, revision weights, and discounted transition expectations receive
truncated integer floors and ceilings at scale 2^{34} . For an expectation, the numerator is $19(3v_1 +$
and the denominator is 100. Its absolute value is checked below 2^{62} before integer division.
In the local randomized lower recursion, multipliers are at most 64 and bounded model values
fit the additional expressions within signed 64-bit capacity. This finite experiment’s overflow
limit is explicit; it is not an arbitrary-horizon machine-arithmetic theorem.

The installed continuous threshold rule can change inside a box. Cost bounds therefore
generate a conservative set of *possible installed actions on the full closed box*. A candidate
action has zero lower revision cost whenever it might equal the installed action; it has zero upper
revision cost only when it is the unique possible installed action. Otherwise the upper revision
weight is charged. This prevents a midpoint installed-action comparison from becoming an
invalid whole-box cost certificate.

The compiled-rule pilot itself chooses the installed action at each box midpoint and then
uses that action throughout the box. It is a proposal, not exactly the continuous installed rule
threshold-crossing boxes. Likewise, the classical solver uses midpoint rewards and bilinear
interpolation only to propose a fixed box policy. Both receive the same whole-box operating and
evaluation. The proof pertains to the resulting serialized box policies, not to an unrecorded
continuous interpolation of their actions.

Backward maxima of lower and upper operating action enclosures bound the unconstrained
operating value. The deterministic necessary-action recursion uses the lower operating witness,
the next-value action bound, and lower implementation costs. The randomized local-grid
recursion instead takes a dual minimum over *all* actions. The code checks both nonnegativity and
the deterministic bound dominates this randomized bound. For a policy with $U_t - J_t^{\text{lower}} \leq$
in every box and date, operating feasibility follows. Its implementation upper enclosure
bounds the feasible optimum from above; a failed operating certificate leaves only a bound
that candidate’s own cost.

A failed sufficient enclosure does not prove that the candidate policy is truly infeasible.
Tables consequently use “certificate pass” rather than treating every failure as a mathe-
matical infeasibility result. The unsuccessful pilot, fallback boxes, coarse-mesh failures, and
candidate-level classical outcomes are retained. Witness construction is charged once for each
step/horizon, while each proposal and reevaluation is charged to its method. This is a nonlinear
state test, not a calibration or a claimed advantage at arbitrary dimension.

7. ALL PRIMARY CONFIGURATIONS

T	Incumbent	ε	Lower	Upper	Gap	Exact
4	defer	0.01	4.936689	4.936689	0.000000	I/P
4	38 defer	0.05	4.932067	4.932067	0.000000	I/P
4	neural31001	0.01	0.111176	0.111176	0.000000	I/P
4	neural31001	0.05	0.074332	0.074332	0.000000	I/P
4	neural31002	0.01	0.149457	0.149457	0.000000	I/P
4	neural31002	0.05	0.112614	0.112614	0.000000	I/P
4	neural31003	0.01	0.121140	0.121140	0.000000	I/P
4	neural31003	0.05	0.084296	0.084296	0.000000	I/P
4	occupancy_stress	0.01	4.206802	4.206802	0.000000	I/P
4	occupancy_stress	0.05	4.189476	4.189476	0.000000	I/P
4	spline129	0.01	0.000000	0.000000	0.000000	I/P
4	spline129	0.05	0.000000	0.000000	0.000000	I/P
4	spline33	0.01	0.000000	0.000000	0.000000	I/P
4	spline33	0.05	0.000000	0.000000	0.000000	I/P
8	defer	0.01	10.810725	10.812347	0.001622	–
8	defer	0.05	10.750714	10.770159	0.019445	–
8	neural31001	0.01	0.468642	0.479631	0.010989	–
8	neural31001	0.05	0.124513	0.227986	0.103473	–
8	neural31002	0.01	0.336417	0.347406	0.010989	–
8	neural31002	0.05	0.044313	0.114248	0.069935	–
8	neural31003	0.01	0.865305	0.871739	0.006434	–
8	neural31003	0.05	0.345015	0.345015	0.000000	I
8	occupancy_stress	0.01	9.490961	9.490961	0.000000	I/P
8	occupancy_stress	0.05	9.248564	9.250944	0.002380	–
8	spline129	0.01	0.000000	0.000000	0.000000	I/P
8	spline129	0.05	0.000000	0.000000	0.000000	I/P
8	spline33	0.01	0.000000	0.000000	0.000000	I/P
8	spline33	0.05	0.000000	0.000000	0.000000	I/P
12	defer	0.01	15.612820	15.614178	0.001359	–
12	defer	0.05	15.471299	15.537400	0.066101	–
12	neural31001	0.01	3.493934	3.508053	0.014119	–
12	neural31001	0.05	2.402580	2.580313	0.177732	–
12	neural31002	0.01	2.536828	2.546365	0.009537	–
12	neural31002	0.05	1.764791	1.840578	0.075787	–
12	neural31003	0.01	1.227595	1.234256	0.006661	–
12	neural31003	0.05	0.140483	0.388955	0.248472	–
12	occupancy_stress	0.01	13.699252	13.702695	0.003443	–
12	occupancy_stress	0.05	13.242542	13.302762	0.060220	–
12	spline129	0.01	0.000000	0.000000	0.000000	I/P
12	spline129	0.05	0.000000	0.000000	0.000000	I/P
12	spline33	0.01	0.000000	0.000000	0.000000	I/P
12	spline33	0.05	0.000000	0.000000	0.000000	I/P

denotes exact initial integrated equality; P additionally denotes all-restart function. Exact fractions and complete hashes are in results/primary_exact_values.json; no equality is used. The machine-readable exact table contains both cost endpoints as strings, the safe-witness lower endpoint, the exact gap, both equality flags, the installed-hash, deployed-policy hash, lower-function hash, and full proof-object hash. In each

TABLE II: All 66 economic sensitivity outcomes

Specification	Rule	ε	Pointwise	Dynamic	Saving
baseline	condition	0.05	10.470918	10.470918	0.000000
baseline	preventive	0.05	10.680893	10.679056	0.001838
baseline	calendar	0.05	8.721989	8.721989	0.000000
baseline	condition	0.20	10.434669	10.434669	0.000000
baseline	preventive	0.20	10.640825	10.630435	0.010390
baseline	calendar	0.20	8.653418	8.653418	0.000000
discount_090	condition	0.05	8.933154	8.933154	0.000000
discount_090	preventive	0.05	9.131155	9.131155	0.000000
discount_090	calendar	0.05	7.458884	7.458884	0.000000
discount_090	condition	0.20	8.906508	8.906508	0.000000
discount_090	preventive	0.20	9.075142	9.074068	0.001074
discount_090	calendar	0.20	7.387071	7.387071	0.000000
discount_099	condition	0.05	11.907352	11.907352	0.000000
discount_099	preventive	0.05	12.107270	12.102384	0.004886
discount_099	calendar	0.05	9.894955	9.894955	0.000000
discount_099	condition	0.20	11.866806	11.866806	0.000000
discount_099	preventive	0.20	12.085757	12.068172	0.017585
discount_099	calendar	0.20	9.842486	9.830059	0.012428
flat_revision_weight	condition	0.05	5.733254	5.733254	0.000000
flat_revision_weight	preventive	0.05	5.898295	5.893513	0.004782
flat_revision_weight	calendar	0.05	4.826153	4.826153	0.000000
flat_revision_weight	condition	0.20	5.733254	5.733254	0.000000
flat_revision_weight	preventive	0.20	5.896158	5.880418	0.015740
flat_revision_weight	calendar	0.20	4.803912	4.803912	0.000000
steep_revision_weight	condition	0.05	15.208582	15.208582	0.000000
steep_revision_weight	preventive	0.05	15.463492	15.462576	0.000916
steep_revision_weight	calendar	0.05	12.617824	12.617824	0.000000
steep_revision_weight	condition	0.20	15.136084	15.136084	0.000000
steep_revision_weight	preventive	0.20	15.385492	15.379338	0.006154
steep_revision_weight	calendar	0.20	12.502923	12.502923	0.000000
high_condition_density	condition	0.05	10.897170	10.897170	0.000000
high_condition_density	preventive	0.05	11.004643	11.003208	0.001435
high_condition_density	calendar	0.05	8.787898	8.787898	0.000000
high_condition_density	condition	0.20	10.870687	10.870687	0.000000
high_condition_density	preventive	0.20	10.973545	10.964602	0.008943
high_condition_density	calendar	0.20	8.703869	8.703869	0.000000
low_condition_density	condition	0.05	10.044666	10.044666	0.000000
low_condition_density	preventive	0.05	10.357144	10.354904	0.002240
low_condition_density	calendar	0.05	8.656080	8.656080	0.000000
low_condition_density	condition	0.20	9.998652	9.998652	0.000000
low_condition_density	preventive	0.20	10.308105	10.296267	0.011838
low_condition_density	calendar	0.20	8.602967	8.602967	0.000000

Specification	Rule	ε	Pointwise	Dynamic	Saving
maintenance_010	condition	0.05	10.432456	10.432456	0.000000
maintenance_010	preventive	0.05	10.633959	10.633959	0.000000
maintenance_010	calendar	0.05	8.677055	8.677055	0.000000
maintenance_010	condition	0.20	10.410489	10.410489	0.000000
maintenance_010	preventive	0.20	10.534915	10.534915	0.000000
maintenance_010	calendar	0.20	8.626888	8.626888	0.000000
maintenance_020	condition	0.05	10.512273	10.512273	0.000000
maintenance_020	preventive	0.05	10.696023	10.696023	0.000000
maintenance_020	calendar	0.05	8.822815	8.772793	0.050022
maintenance_020	condition	0.20	10.068605	10.055233	0.013372
maintenance_020	preventive	0.20	10.577155	10.577155	0.000000
maintenance_020	calendar	0.20	8.257042	8.170031	0.087011
defer_persistence_065	condition	0.05	10.469741	10.469741	0.000000
defer_persistence_065	preventive	0.05	10.679481	10.678541	0.000940
defer_persistence_065	calendar	0.05	8.721618	8.721618	0.000000
defer_persistence_065	condition	0.20	10.434168	10.434168	0.000000
defer_persistence_065	preventive	0.20	10.640244	10.630954	0.009290
defer_persistence_065	calendar	0.20	8.653691	8.653691	0.000000
defer_persistence_085	condition	0.05	8.309909	8.309909	0.000000
defer_persistence_085	preventive	0.05	9.281022	9.281022	0.000000
defer_persistence_085	calendar	0.05	6.696913	6.696913	0.000000
defer_persistence_085	condition	0.20	7.808322	7.792568	0.015754
defer_persistence_085	preventive	0.20	8.734932	8.692720	0.042212
defer_persistence_085	calendar	0.20	6.142199	6.131385	0.010814

every displayed case passed the same-class operating check for both policies. All values are normalized discounted implementation costs; exact rational values, changed primitives, and hashes are in sensitivity.json and its proof objects.

9. ALL FIXED-RESTART CONSTRAINED REFERENCES

TABLE III: All exact fixed-restart history-conditioned references

Incumbent	ε	x_0	Randomized	Det. lower	Nodes	Vertices
defer	0.01	0	4.149907	4.365233	253	314
defer	0.01	1/4	4.447063	4.658161	257	318
defer	0.01	1/2	4.743143	4.951089	253	314
defer	0.01	3/4	5.032875	5.244017	256	316
defer	0.01	1	5.422816	5.615933	212	264
defer	0.05	0	3.287636	4.365233	253	314
defer	0.05	1/4	3.600854	4.658161	257	318
defer	0.05	1/2	3.909072	4.951089	253	314
defer	0.05	3/4	4.188360	5.244017	256	316
defer	0.05	1	4.647604	5.615933	212	264
neural31001	0.01	0	0.000000	0.000000	253	253
neural31001	0.01	1/4	0.000000	0.000000	257	258
neural31001	0.01	1/2	0.000000	0.000000	253	254

Incumbent	ε	x_0	Randomized	Det. lower	Nodes	Vertices
neural31001	0.01	3/4	1.467067	1.750000	256	260
neural31001	0.01	1	0.000000	0.000000	212	212
neural31001	0.05	0	0.000000	0.000000	253	253
neural31001	0.05	1/4	0.000000	0.000000	257	258
neural31001	0.05	1/2	0.000000	0.000000	253	254
neural31001	0.05	3/4	0.335335	1.750000	256	260
neural31001	0.05	1	0.000000	0.000000	212	212
neural31002	0.01	0	0.000000	0.000000	253	253
neural31002	0.01	1/4	0.000000	0.000000	257	257
neural31002	0.01	1/2	0.000000	0.000000	253	253
neural31002	0.01	3/4	1.467067	1.750000	256	258
neural31002	0.01	1	0.000000	0.000000	212	212
neural31002	0.05	0	0.000000	0.000000	253	253
neural31002	0.05	1/4	0.000000	0.000000	257	257
neural31002	0.05	1/2	0.000000	0.000000	253	253
neural31002	0.05	3/4	0.335335	1.750000	256	258
neural31002	0.05	1	0.000000	0.000000	212	212
neural31003	0.01	0	0.000000	0.000000	253	253
neural31003	0.01	1/4	0.000000	0.000000	257	258
neural31003	0.01	1/2	0.000000	0.000000	253	254
neural31003	0.01	3/4	1.467067	1.750000	256	261
neural31003	0.01	1	0.000000	0.000000	212	213
neural31003	0.05	0	0.000000	0.000000	253	253
neural31003	0.05	1/4	0.000000	0.000000	257	258
neural31003	0.05	1/2	0.000000	0.000000	253	254
neural31003	0.05	3/4	0.335335	1.750000	256	261
neural31003	0.05	1	0.000000	0.000000	212	213
occupancy_stress	0.01	0	4.149907	4.365233	253	312
occupancy_stress	0.01	1/4	4.447063	4.658161	257	316
occupancy_stress	0.01	1/2	3.774393	3.954007	253	303
occupancy_stress	0.01	3/4	4.036576	4.234366	256	302
occupancy_stress	0.01	1	0.000000	0.000000	212	237
occupancy_stress	0.05	0	3.287636	4.365233	253	312
occupancy_stress	0.05	1/4	3.600854	4.658161	257	316
occupancy_stress	0.05	1/2	3.051087	3.954007	253	303
occupancy_stress	0.05	3/4	3.252671	4.234366	256	302
occupancy_stress	0.05	1	0.000000	0.000000	212	239
spline129	0.01	0	0.000000	0.000000	253	253
spline129	0.01	1/4	0.000000	0.000000	257	257
spline129	0.01	1/2	0.000000	0.000000	253	253
spline129	0.01	3/4	0.000000	0.000000	256	256
spline129	0.01	1	0.000000	0.000000	212	212
spline129	0.05	0	0.000000	0.000000	253	253
spline129	0.05	1/4	0.000000	0.000000	257	257
spline129	0.05	1/2	0.000000	0.000000	253	253
spline129	0.05	3/4	0.000000	0.000000	256	256
spline129	0.05	1	0.000000	0.000000	212	212
spline33	0.01	0	0.000000	0.000000	253	253
spline33	0.01	1/4	0.000000	0.000000	257	257
spline33	0.01	1/2	0.000000	0.000000	253	253

Incumbent	ε	x_0	Randomized	Det. lower	Nodes	Vertices
spline33	0.01	3/4	0.000000	0.000000	256	256
spline33	0.01	1	0.000000	0.000000	212	212
spline33	0.05	0	0.000000	0.000000	253	253
spline33	0.05	1/4	0.000000	0.000000	257	257
spline33	0.05	1/2	0.000000	0.000000	253	253
spline33	0.05	3/4	0.000000	0.000000	256	256
spline33	0.05	1	0.000000	0.000000	212	212

each objective is a Dirac restart, not the uniform-initial Markov objective. Exact rational frontier nodes, and independent support checks are in the corresponding frontiers proof

10. ALL NONLINEAR CANDIDATES AND FAILURES

TABLE IV: All nonlinear candidates, including failed certificates

T	N	Method	Pass	Regret bound	Global LB	Policy UB	Seconds
4	16	R0	yes	0.4977	0.0000	2.6751	0.011
4	16	R4	yes	0.4977	0.0000	2.4974	0.010
4	16	P	no	1.1074	0.0000	1.6213	0.007
4	16	C	yes	0.4151	0.0000	3.0150	0.024
4	32	R0	no	0.5157	0.0000	2.1002	0.012
4	32	R4	yes	0.4996	0.0000	2.1219	0.012
4	32	P	no	0.9009	0.0000	0.6582	0.012
4	32	C	yes	0.3095	0.0000	2.6452	0.038
4	64	R0	yes	0.5000	0.0000	1.7191	0.030
4	64	R4	yes	0.4996	0.0000	1.9953	0.031
4	64	P	no	0.8199	0.0000	0.3489	0.028
4	64	C	yes	0.2182	0.0000	2.6183	0.083
4	128	R0	yes	0.5000	0.0000	1.3502	0.103
4	128	R4	yes	0.5000	0.0000	1.5348	0.105
4	128	P	no	0.7748	0.0000	0.1618	0.100
4	128	C	yes	0.1651	0.0000	2.5997	0.279
4	256	R0	yes	0.5000	0.0000	0.9991	0.369
4	256	R4	yes	0.5000	0.0000	1.1250	0.377
4	256	C	yes	0.1539	0.0000	2.5915	1.049
8	16	R0	no	0.7217	0.0000	7.8440	0.014
8	16	R4	no	0.7207	0.0000	7.5944	0.014
8	16	P	no	1.9218	0.0000	3.5345	0.014
8	16	C	no	1.8675	0.0000	3.2901	0.048
8	32	R0	no	0.5306	0.0000	5.3481	0.024
8	32	R4	yes	0.4999	0.0000	5.1263	0.024
8	32	P	no	1.4938	0.0000	1.0518	0.023
8	32	C	yes	0.4683	0.0000	5.2667	0.075
8	64	R0	yes	0.5000	0.0000	4.5263	0.060
8	64	R4	yes	0.4998	0.0000	3.9348	0.060
8	64	P	no	1.3586	0.0000	0.5982	0.057

T	N	Method	Pass	Regret bound	Global LB	Policy UB	Seconds
8	64	C	yes	0.2950	0.0000	5.2035	0.170
8	128	R0	yes	0.5000	0.0000	4.1115	0.209
8	128	R4	yes	0.5000	0.0000	5.0151	0.207
8	128	P	no	1.2699	0.0000	0.1830	0.191
8	128	C	yes	0.1980	0.0000	5.1793	0.569
8	256	R0	yes	0.5000	0.0000	3.0417	0.720
8	256	R4	yes	0.5000	0.0000	3.0816	0.722
8	256	C	yes	0.1591	0.0000	5.1651	2.051
16	16	R0	no	1.1505	0.0000	12.7079	0.028
16	16	R4	no	1.1499	0.0000	12.5424	0.028
16	16	P	no	3.0806	0.0000	6.0818	0.027
16	16	C	no	3.0440	0.0000	5.9726	0.095
16	32	R0	no	0.6609	0.0000	13.1739	0.047
16	32	R4	no	0.6250	0.0000	12.3093	0.046
16	32	P	no	2.3354	0.0000	1.6657	0.045
16	32	C	no	2.3030	0.0000	1.8973	0.148
16	64	R0	yes	0.5000	0.0000	8.7753	0.124
16	64	R4	yes	0.5000	0.0000	8.7555	0.119
16	64	P	no	2.0828	0.0000	1.1280	0.107
16	64	C	yes	0.4225	0.0000	9.0306	0.348
16	128	R0	yes	0.5000	0.0000	7.9683	0.400
16	128	R4	yes	0.5000	0.0000	7.6643	0.393
16	128	P	no	1.9345	0.0000	0.2580	0.372
16	128	C	yes	0.2597	0.0000	8.9885	1.118
16	256	R0	yes	0.5000	0.0000	7.3118	1.436
16	256	R4	yes	0.5000	0.0000	7.4019	1.430
16	256	C	yes	0.1931	0.0000	8.9669	4.103
32	16	R0	no	1.6264	0.0000	17.9085	0.055
32	16	R4	no	1.6261	0.0000	17.8356	0.056
32	16	P	no	4.3523	0.0000	8.9351	0.054
32	16	C	no	4.2975	0.0000	8.8716	0.188
32	32	R0	no	0.8575	0.0000	24.2374	0.091
32	32	R4	no	0.8417	0.0000	23.5439	0.094
32	32	P	no	3.2834	0.0000	2.3702	0.087
32	32	C	no	3.2552	0.0000	2.6072	0.293
32	64	R0	no	0.5056	0.0000	16.2448	0.242
32	64	R4	no	0.5013	0.0000	14.0445	0.238
32	64	P	no	2.8959	0.0000	1.7564	0.211
32	64	C	yes	0.4440	0.0000	16.7883	0.684
32	128	R0	yes	0.5000	0.0000	12.5711	0.797
32	128	R4	yes	0.5000	0.0000	12.5225	0.832
32	128	P	no	2.6814	0.0000	0.3477	0.705
32	128	C	yes	0.3209	0.0000	13.1873	2.276
32	256	R0	yes	0.5000	0.0000	11.8492	2.867
32	256	R4	yes	0.5000	0.0000	12.1034	2.876
32	256	C	yes	0.2305	0.0000	13.1578	8.256

es. R0/R4 are restart generators; C is the classical four-price portfolio; P is the midpoint-piled installed-rule pilot. Global LB is the deterministic outer-class lower bound. Policy UB

TABLE V: All adaptive local LP certificates

Witness	ε	ρ	Grid LB	Adaptive LB	Adaptive UB	Cells
constructed	0.01	1/100	0.076185	0.088078	0.106627	3159
constructed	0.01	1/200	0.076185	0.088107	0.097381	6275
constructed	0.01	1/400	0.076185	0.088121	0.092758	12529
exact	0.01	1/100	0.076186	0.088078	0.106627	3156
exact	0.01	1/200	0.076186	0.088107	0.097382	6272
exact	0.01	1/400	0.076186	0.088121	0.092759	12526
constructed	0.05	1/100	0.020289	0.025619	0.044169	171
constructed	0.05	1/200	0.020289	0.025649	0.034924	309
constructed	0.05	1/400	0.020289	0.025665	0.030302	588
exact	0.05	1/100	0.020289	0.025620	0.044169	168
exact	0.05	1/200	0.020289	0.025649	0.034924	306
exact	0.05	1/400	0.020289	0.025665	0.030302	585

every accepted open cell passes exact feasibility and quadratic gap tests; stored isolated cells have exact point-LP solutions. The floor is zero in all rows.

12. KILLED OPERATORS: ERROR PROPAGATION AND ACTION COVERAGE

an exact killed step as $T_t^a v = h_t^a + Q_t^a v$, where Q_t^a is a positive subprobability kernel with mass bounded by α_t . Monotonicity and the estimate $\|T_t^a v - T_t^a w\|_\infty \leq \alpha_t \|v - w\|_\infty$ follow directly. Suppose a validated approximation obeys $\|\tilde{T}_t^a v - T_t^a v\|_\infty \leq \xi_t$ for every action a in the support of the policy function used by the construction. If upper encoding adds defect d_t , then the backward error obeys

$$b_t = d_t + \xi_t + \alpha_t b_{t+1}.$$

an upper encoded step by ξ_t and taking the corresponding lower envelope supplies safe lower bounds. The same positive-kernel argument proves the deterministic necessary-action lower bound for an implementation recursion with compatible stopping and nonnegative costs. A quantifier over actions is essential. A discrete subset gives a lower operating value but not the true optimum. It also creates an overly narrow implementation policy class. To certify a certificate for the continuous-control target, one must enclose all omitted actions, for all actions with validated action regularity and interval covers. State interpolation, killed-transition approximation, action coverage, and time-discretization error must remain separately identifiable components of the budget. The present nonlinear maintenance calculation validates a distinct finite-action approximation, not these diffusion approximation constants.

13. RETAINED SCALAR STOPPED-CONTROL OPTIMUM

start $(0, 61/50, 5/4)$ with $k = 2$, $c = 3/4$, $p = 0$, and constant $\theta \in [-1/5, 1/5]$, the retained stopped-economy payoff is strictly increasing in θ . The derivative lower bounds

13.1. Deterministic wealth and the Dynkin integrand

Throughout this section the primitives, controls, and restart are those stated immediately above. With $p = 0$, $c = 3/4$, and $x(0) = 5/4$,

$$x(t) = \frac{75}{2} - \frac{145}{4}e^{t/50}. \quad (1)$$

For $0 \leq t \leq 1$, $e^{t/50} \leq (1 - 1/50)^{-1} = 50/49$, hence $x(t) \geq 25/49 > 1/2$. It is decreasing and starts below two, so no wealth boundary is reached. Let y denote preference. Direct differentiation of the unchanged settlement gives

$$h_\theta(t, y) = \frac{(3/4)^{1-y}}{1-y} - \theta^2 + 8 - .04\theta(y-2) - .00005 + .002 - \frac{.075}{x(t)} + .0008(y-2)^2 - .004 \log x(t) + .32(1-t), \quad (2)$$

$$\partial_\theta h_\theta = -2\theta - .04(y-2). \quad (3)$$

For $y \in [6/5, 14/5]$, put $z = y - 1 \in [1/5, 9/5]$ and $L = \log(4/3) < 1/3$. The utility is $-e^{Lz}/z$, which is increasing on this interval because $Lz < 1$. Its lower bound is $-5e^{1/15} \geq -75/14$. For $|y-2| \leq 4/5$, $x(t) \geq 1/2$, and $\log x(t) \leq \log(5/4) < 1/4$. Dropping nonnegative terms yields

$$h_\theta \geq 8 - \frac{1}{20000} - \frac{3}{20} - \frac{1}{1000} - \frac{4}{625} - \frac{1}{25} - \frac{75}{14} = \frac{342357}{140000} > 2. \quad (4)$$

The utility derivative is $e^{Lz}(1 - Lz)/z^2 \geq 10/81$. Thus

$$\partial_y h_\theta \geq \frac{10}{81} - \frac{4}{3125} - \frac{1}{125} = \frac{28901}{253125} > 0. \quad (5)$$

and $h_\theta(t, y)$ constantly from $b = 14/5$ to larger y . The extension remains bounded, nondecreasing in y , and at least two. Its parameter derivative retains the bounds in (3). Conservative absolute bounds $|h| \leq 20$ and $|\partial_\theta h| \leq 2$ hold for this extension.

13.2. A coupling lower bound for the derivative

Let $Y_t = u_0 + \theta t + \sigma W_t$, killed at $a = 6/5$, where $u_0 = 61/50$ and $\sigma = 1/20$. If drift increases, the coupled process is higher at every date and lower-boundary survival events are nested. For fixed nondecreasing integrand $h \geq 2$, paths surviving under both drifts have a nonnegative off change, while newly surviving paths contribute at least two. Taking a positive difference quotient gives

$$\partial_\theta \mathbb{E}_\theta[h(t, Y_t) \mathbf{1}\{t < \tau_a\}] \geq 2\partial_\theta \Pr_\theta(t < \tau_a). \quad (6)$$

The direct dependence of h_θ on θ must be added, not included in this frozen-integrand comparison. It contributes at least $-4/125$ for $\theta \in [-1/5, 0]$ and at least $-54/125$ for $\theta \in [0, 1/5]$, after

stability and dominated passage to the occupation integral for this bounded extension. Reflection formula with drift, differentiated exactly, yields

$$46 \quad S'_\theta(t) = 16e^{-16\theta} \Phi\left(\frac{\theta t - 1/50}{(1/20)\sqrt{t}}\right). \quad (7)$$

tion $\log \Phi$ is concave. Indeed, with $r = \varphi/\Phi$, its second derivative is $-r(z)\{z + r(z)\} < 0$ for $z < 0$, the elementary Gaussian tail inequality gives $r(z) > -z$. Equation (7) is therefore concave in θ . Its minimum on each of the two half intervals is at least the smaller of its endpoint values.

On $[-1/5, 0]$, the normal arguments at $\theta = 0$ and $\theta = -1/5$ are bounded below by $-2\sqrt{t}$ and $-4\sqrt{t}$, respectively. The latter follows by maximizing $4\sqrt{t} + (2/5)/\sqrt{t}$ on that compact interval. Its maximum is $14/5$. The rational inequalities

$$24 < e^{16/5} < 25, \quad \Phi(-2) > 1/50, \quad \Phi(-14/5) > 1/500, \quad (8)$$

$$\Phi(-4/5) > 1/5, \quad \Phi(6/5) > 22/25.$$

On $[0, 1/5]$, $S'_\theta(t) \geq 8/25$ on the entire negative drift half interval.

On $[1/5, 1]$, the endpoint arguments at $\theta = 0$ and $\theta = 1/5$ are at least $-4/5$ and $6/5$. On $[0, 1/5]$, $S''_\theta(t) \geq 352/625$ on the positive drift half interval. Since $e^{-.04t} \geq 24/25$, the two lower bounds for the derivative of the lower-killed occupation expression are

$$2\frac{4}{25}\frac{24}{25}\frac{8}{25} - \frac{4}{125} = .066304, \quad (9)$$

$$2\frac{3}{4}\frac{24}{25}\frac{352}{625} - \frac{54}{125} = .379008. \quad (10)$$

A discrete sample grid appears in this argument. The intervals cover every admissible constant drift value including the shared point zero.

13.3. Distant-boundary error and rational checks

The upper boundary is distant but not discarded. Under all admissible drifts, a path hitting it has a Brownian excursion of at least $(b - u_0 - 1/5)/\sigma$. Reflection and a Gaussian tail bound give

$$p_b \leq 2 \exp\left[-\frac{(b - u_0 - 1/5)^2}{2\sigma^2}\right] = 2e^{-380.88} < 2^{-379}. \quad (11)$$

With this b -dependent difference supported on that event, the drift likelihood score has second moment at most $1/\sigma^2$. Cauchy–Schwarz bounds the derivative discrepancy between the actual lower-killed occupation integral and the extended lower-killed one by

$$\frac{20}{\sigma} \sqrt{p_b} + 2p_b < 400 \cdot 2^{-189} + 2 \cdot 2^{-379} < 1/1000. \quad (12)$$

Adding $1/1000$ from the preceding half-line bounds gives precisely $8163/125000$ and $8163/125000$. A strictly increasing continuous function on the compact drift interval has the maximizer $1/5$.

elementary constants in (8) with exact fractions. Exponential bounds use Taylor sums and an explicit geometric tail. A Machin arctangent identity with alternating-series bounds proves a rational bound for π , and $\sqrt{2\pi} < 251/100$ follows. Gaussian lower bounds use monotone right sums of mesh $1/100$ on finite intervals, while exponential bounds are bounded by a rational Taylor approximation. Truncating the infinite tail only lowers a tail integral. For $\Phi(6/5)$ the integral is approximated from zero to $6/5$, added to $1/2$. Terms are rounded downward to a common dyadic denominator for summation, with the direction of the bound preserved. The code records the resulting rational lower bounds and requires them to exceed all displayed thresholds. The payoff enclosure at $\theta = 1/5$ is inherited from the unchanged validated R30 killed-kernel model. Its source path, Git blob hash, and the fact that it is not recomputed are recorded in `revisions/2026-09-24-r34/results/boundary.json`. The current theorem changes its interpretation from an endpoint evaluation to the optimum of the entire constant region. It does not assert feedback optimality or satisfy the original all-domain current-state target.

14. PRESERVATION MAP AND HISTORICAL EVIDENCE

No earlier revision source, result, or review is removed. The branch starts at the latest R36 new commit and adds R38. The article, supplement, response, and computational report from R36 are preserved page-for-page in `HISTORY_R38.pdf`, followed by the earlier recursive historical annex. The original stopped consumption–portfolio specification, its full-domain target, scalar boundary quadrature record, the 47-coordinate derivative/residual bridge discussion, firm transport and rollback derivations, inventory comparisons, and operating-mode studies remain in those sources and archived pages.

The R35 and R36 protocol files remain unchanged in the repository. Their unmaterialized execution assertions are not treated as verified observations. R38 supplies a newly executed scientific object with explicit provenance. In particular, the fresh compiled-rule nonlinear pilot is not relabeled as a historical zero-penalty run. The complete prior unfavorable neural and inventory results are retained; none is converted into a new neural acceleration claim.

The old material is a historical annex, not additional experiments for the R38 cohort. Its former elements of current revision status are read at their original dates. The current main article and response specify which numerical objectives have new certificates and which original targets remain in their previous bounds. This organization shortens the main narrative without deleting the technical record. The main article prints the references cited in its current theorem and verification discussion. The complete inherited bibliography is retained in `history/references_R34.tex`, in the unchanged R34 source, and in the historical PDF annex; uncited predecessor references have not been removed from the scientific archive.